AWS Enterprise Containerized App, Governance & Multi-Account Security

Estimated Duration: 1.5 to 2 Hours
Focus Areas: Amazon ECR, Amazon ECS (Fargate), AWS Application Load Balancer, AWS Organizations, Service Control Policies (SCPs), AWS Security Hub, AWS Config

Architecture Overview & Blueprint

This hands-on lab covers essential container orchestration and multi-account security governance concepts tested on the AWS Solutions Architect Associate (SAA-C03) exam: deploying serverless microservices using Amazon ECS on AWS Fargate behind an Application Load Balancer, combined with centralized policy enforcement, compliance tracking, and vulnerability auditing.

Component	AWS Service	Architectural Purpose
Container Registry	Amazon ECR	Stores versioned Docker container images securely with image scanning on push.
Serverless Compute Tier	Amazon ECS (Fargate)	Executes containerized workloads statelessly without managing underlying EC2 instances.
Ingress & Routing	Application Load Balancer (ALB)	Distributes incoming HTTP/HTTPS traffic across ECS task targets in multiple Availability Zones.
Organization Governance	AWS Organizations & SCPs	Enforces preventive guardrails and restricts unapproved actions across accounts.
Compliance & Auditing	AWS Config & Security Hub	Monitors resource configurations continuously and

Component	AWS Service	Architectural Purpose
		centralizes security standard assessments.

Description of the Project

In this lab, you will design and implement a secure containerized application environment paired with enterprise compliance guardrails.

- **Container Repository & Build:** Create an Amazon ECR repository with automated image scanning enabled, build a microservice container image, and push it to ECR.
- **Serverless Orchestration & Load Balancing:** Define an ECS Task Definition with appropriate vCPU/memory allocations and IAM roles, then deploy an ECS Service on AWS Fargate integrated with an Application Load Balancer across multi-AZ public/private subnets.
- **Preventive Governance (SCPs):** Configure Service Control Policies (SCPs) in AWS Organizations to restrict resource creation in unauthorized regions and prevent non-compliant infrastructure changes.
- **Detective Security & Compliance:** Enable AWS Config rules to monitor EBS encryption and Security Group ingress rules, and deploy AWS Security Hub to assess overall compliance against the CIS AWS Foundations Benchmark.

Step-by-Step Implementation Guide

Step 1: Create Amazon ECR Repository & Push Container Image

- Navigate to **Amazon ECR Console** → **Repositories** → **Create repository**.
- **Repository Name:** `enterprise-microservice-app`
- **Tag Immutability:** Enable | **Scan on Push:** Enable
- Build and push a lightweight Docker container image using AWS CloudShell or local CLI commands:
  - Log in to your ECR registry via AWS CLI using the `ecr get-login-password` command.
  - Build the Docker image locally: `docker build -t enterprise-microservice-app .`
  - Tag the image with your ECR repository URI.
  - Push the image to Amazon ECR.

Step 2: Configure ALB Target Group & Application Load Balancer

- Navigate to **EC2 Console** → **Target Groups** → **Create Target Group**.

- **Target Type:** IP addresses (Required for ECS Fargate) | **Protocol:** HTTP | **Port:** 80
- **Health Check Path:** `/`
- Navigate to **Load Balancers** → **Create Application Load Balancer**.
- **Name:** `ecs-app-alb` | **Scheme:** Internet-facing
- Select VPC and assign public subnets across 2 Availability Zones.
- Attach a Security Group allowing inbound HTTP (Port 80) traffic.
- Point HTTP Listener to the created Target Group.

Step 3: Deploy ECS Cluster, Task Definition & Fargate Service

- Navigate to **Amazon ECS Console** → **Task Definitions** → **Create new Task Definition**.
- **Launch Type:** AWS Fargate | **OS/Architecture:** Linux/X86_64
- **Memory:** 0.5 GB | **CPU:** 0.25 vCPU
- **Container Definition:** Name = `web-app`, Image URI = *your ECR image URI*, Port Mapping = 80 TCP
- **Execution Role:** `ecsTaskExecutionRole` (with `AmazonECSTaskExecutionRolePolicy` attached)
- **Create ECS Cluster:** Name = `enterprise-ecs-cluster`
- **Create ECS Service:**
  - **Launch Type:** Fargate | **Deployment option:** Service
  - **Task Definition:** Select `web-app` | **Desired tasks:** 2
  - **Networking:** Select VPC and private subnets
  - **Load Balancing:** Select existing ALB (`ecs-app-alb`) and Target Group
- Verify that ECS tasks launch successfully and pass ALB target health checks.

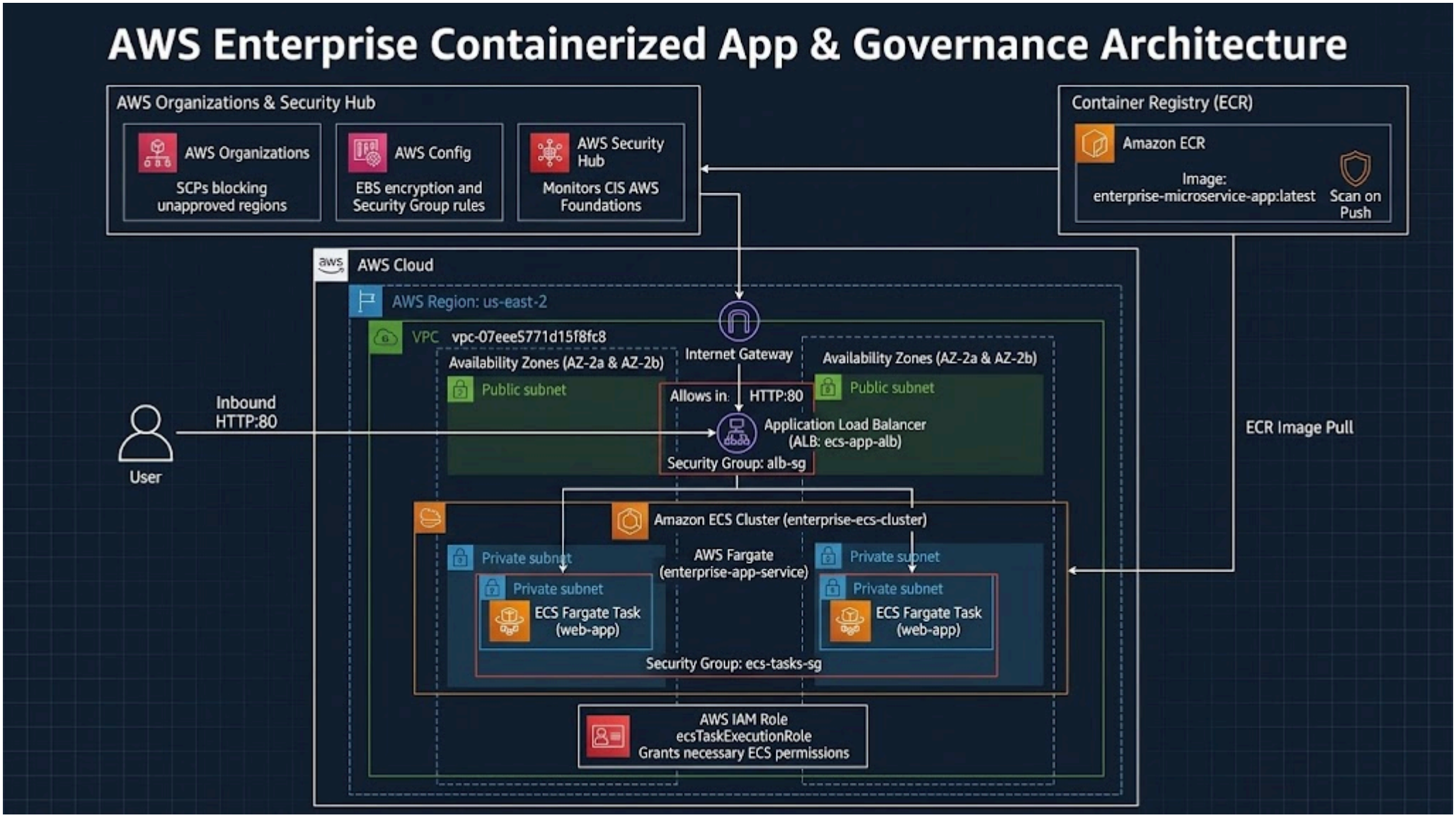
Step 4: Implement AWS Organizations Service Control Policies (SCPs)

- Navigate to **AWS Organizations Console** → **Policies** → **Service Control Policies**.
- Click **Create policy**:
  - **Policy Name:** `DenyUnapprovedRegionsAndUnencryptedEBS`
  - Add policy statement restricting actions outside permitted regions (e.g., `us-east-1`, `us-west-2`) except for global services like IAM, Route53, and CloudFront.
- Attach the policy to target Organizational Units (OUs) or Member Accounts.

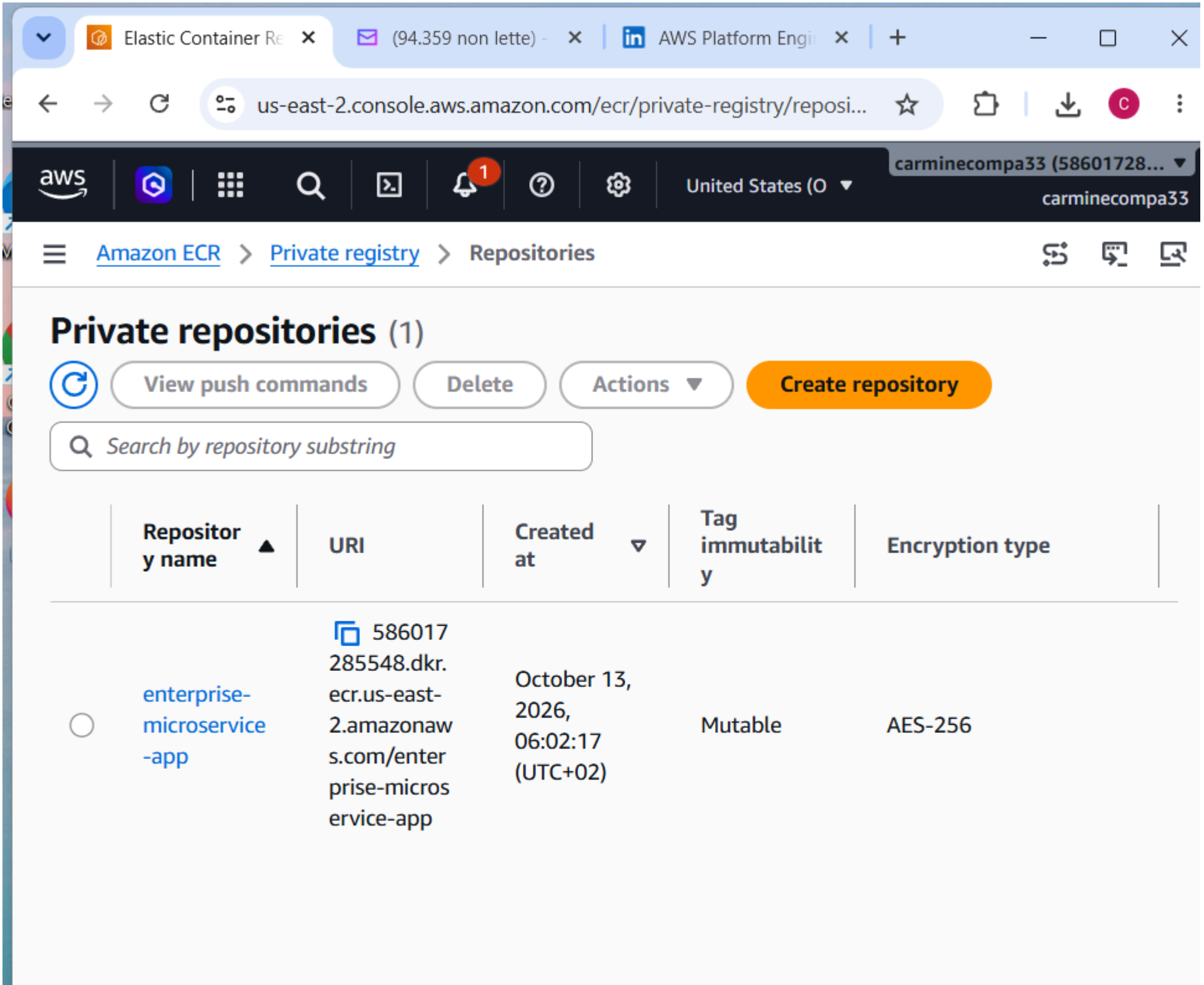
Step 5: Enable AWS Config & Security Hub Compliance Monitoring

- Navigate to **AWS Config Console** → **Settings** → **Turn on recording**.
- Enable Managed Rules:
  - `encrypted-volumes-check`: Checks whether EBS volumes attached to instances are encrypted.

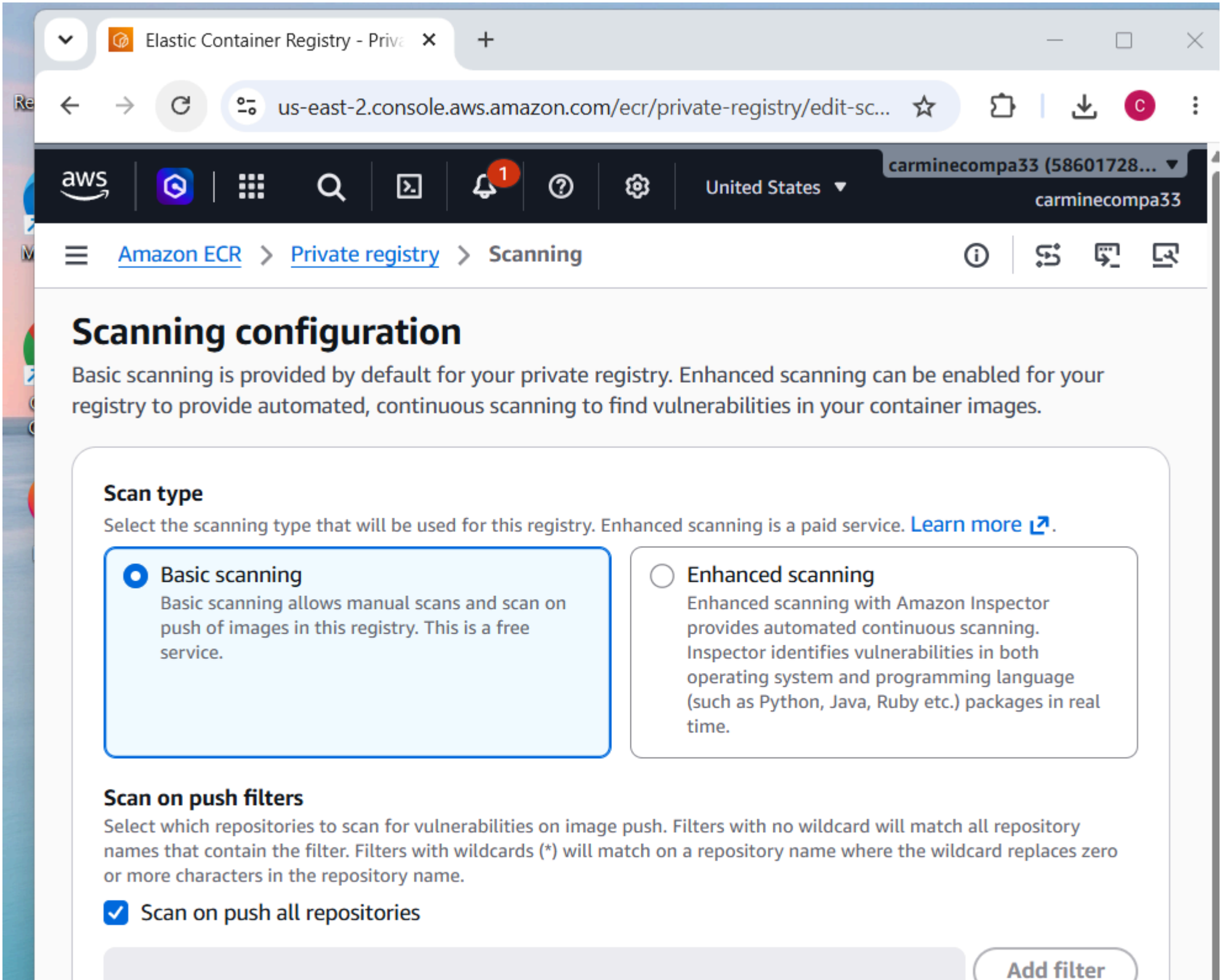
    - `vpc-sg-open-only-to-authorized-ports`: Checks whether Security Groups disallow unrestricted inbound access (`0.0.0.0/0`) on port 22/3389.
- Navigate to **AWS Security Hub Console** → **Enable Security Hub**.
- Enable standard standards: **CIS AWS Foundations Benchmark v1.4.0** and **AWS Foundational Security Best Practices**.
- Inspect compliance score dashboard and review finding recommendations for remediating non-compliant resources.



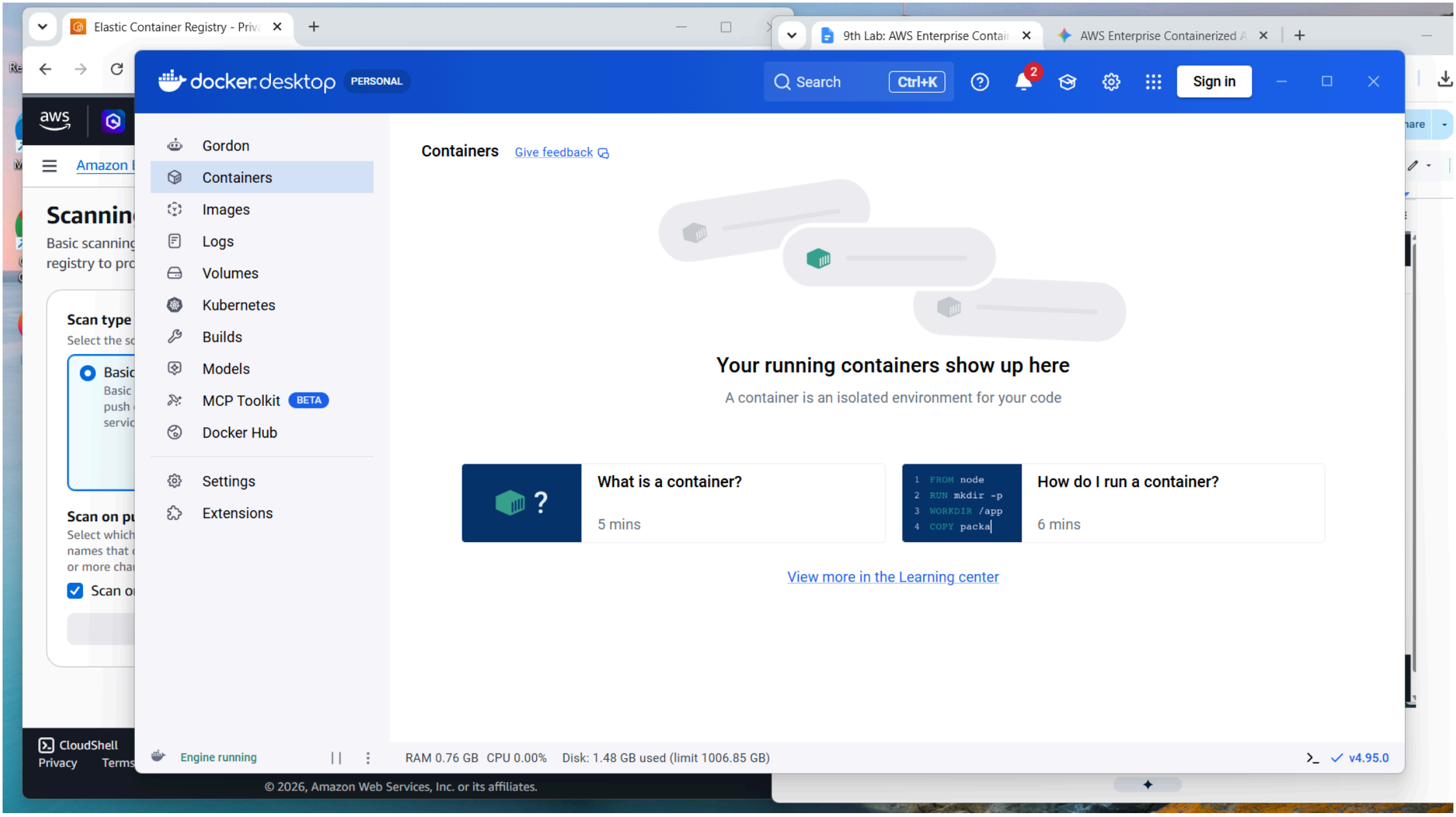
Created a private repository in AWS Elastic Container Registry


Elastic Container Re
(94.359 non lette) -
AWS Platform Engi
us-east-2.console.aws.amazon.com/ecr/private-registry/reposi...
aws
United States (O
carminecompa33 (58601728...
carminecompa33
Amazon ECR
Private registry
Repositories
Private repositories (1)
View push commands
Delete
Actions
Create repository
Search by repository substring
Repository name
URI
Created at
Tag immutability
Encryption type
enterprise-microservice-app
586017285548.dkr.ecr.us-east-2.amazonaws.com/enterprise-microservice-app
October 13, 2026, 06:02:17 (UTC+02)
Mutable
AES-256

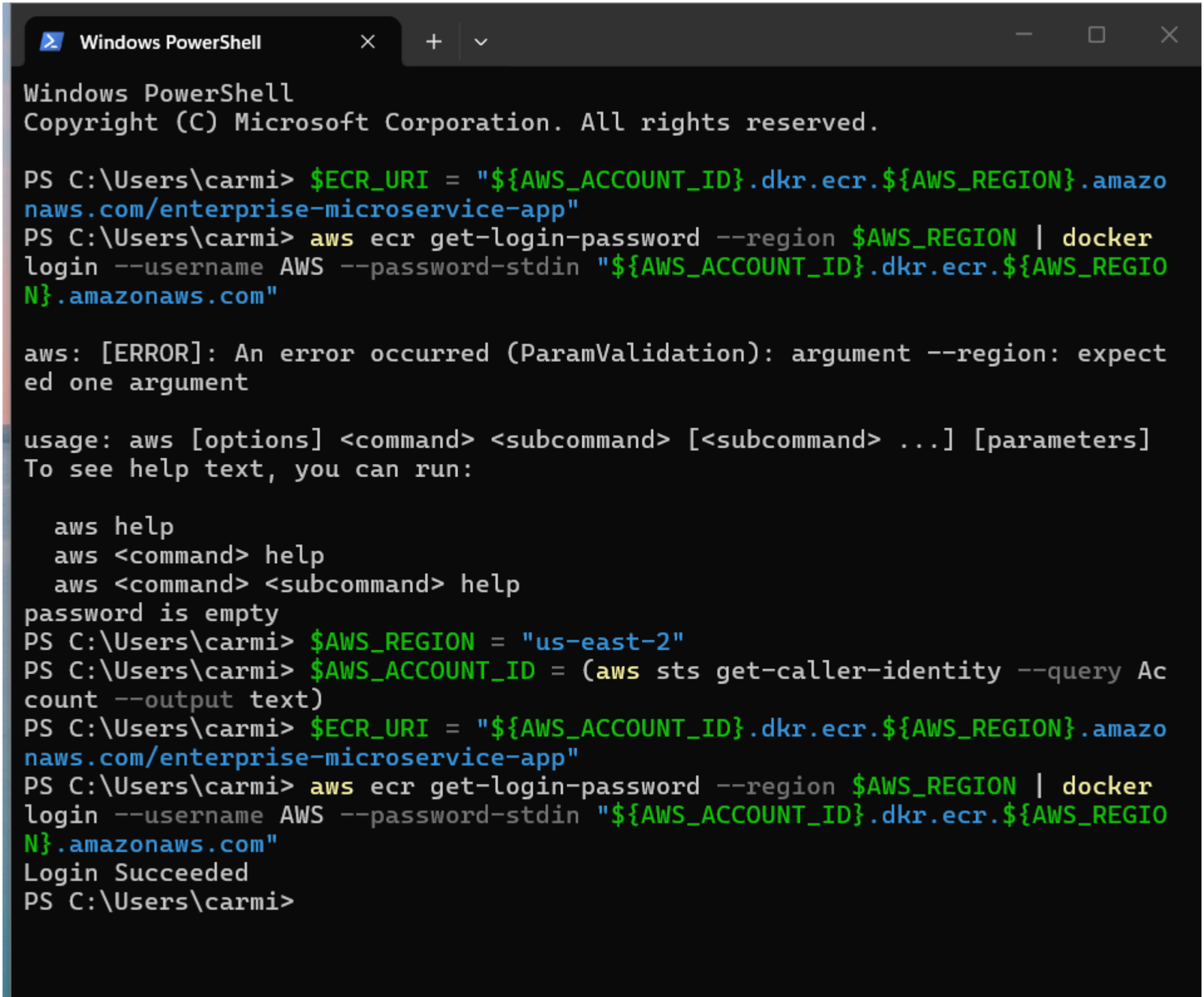
Enable Basic scanning on push for all repositories


Elastic Container Registry - Priva
us-east-2.console.aws.amazon.com/ecr/private-registry/edit-sc...
aws
United States
carminecompa33 (58601728...
carminecompa33
Amazon ECR
Private registry
Scanning
Scanning configuration
Basic scanning is provided by default for your private registry. Enhanced scanning can be enabled for your registry to provide automated, continuous scanning to find vulnerabilities in your container images.
Scan type
Select the scanning type that will be used for this registry. Enhanced scanning is a paid service. Learn more.
Basic scanning
Basic scanning allows manual scans and scan on push of images in this registry. This is a free service.
Enhanced scanning
Enhanced scanning with Amazon Inspector provides automated continuous scanning. Inspector identifies vulnerabilities in both operating system and programming language (such as Python, Java, Ruby etc.) packages in real time.
Scan on push filters
Select which repositories to scan for vulnerabilities on image push. Filters with no wildcard will match all repository names that contain the filter. Filters with wildcards (*) will match on a repository name where the wildcard replaces zero or more characters in the repository name.
Scan on push all repositories
Add filter

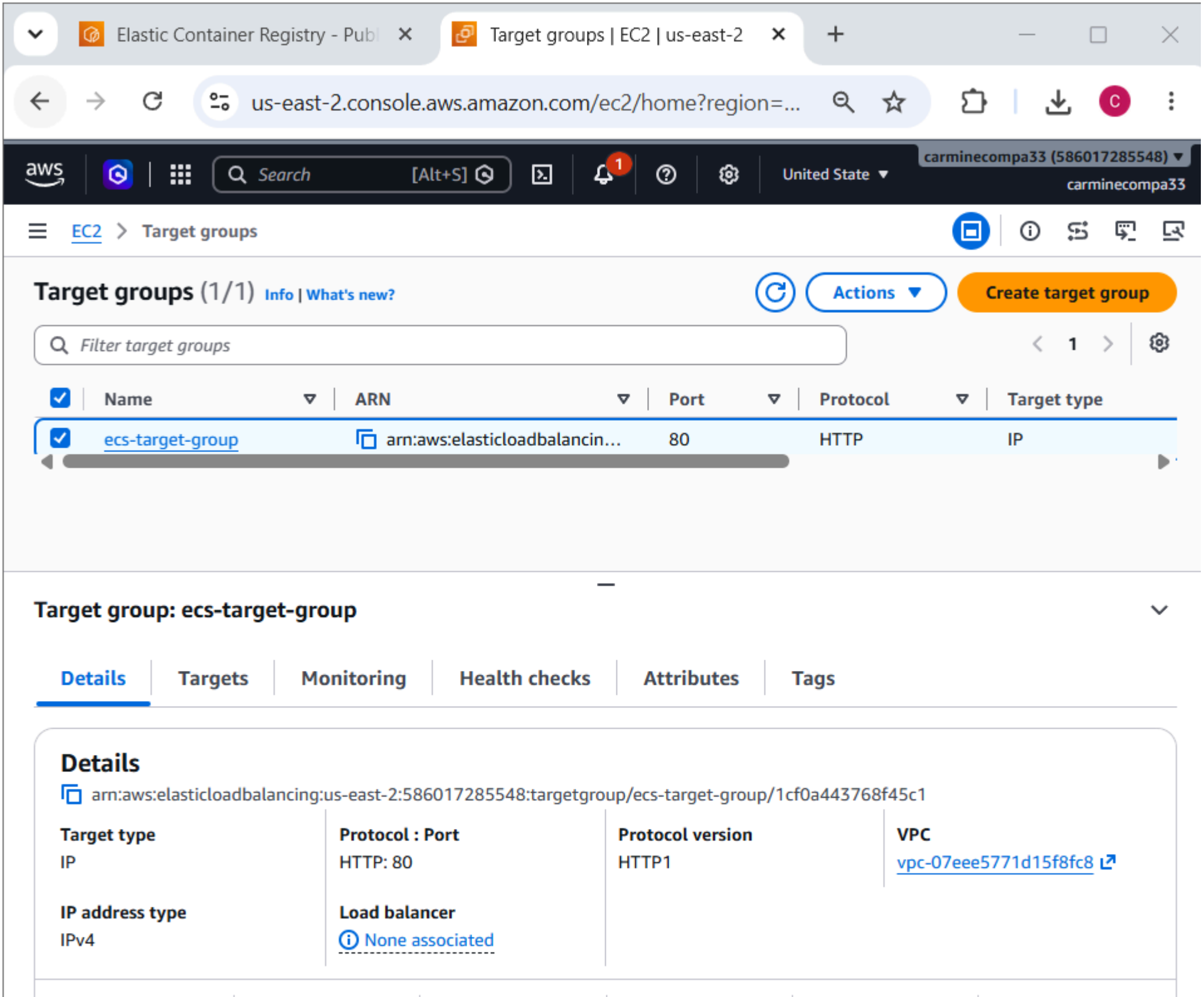
Installed Docker on my local machine



Login into my AWS ECR registry via AWS Docker CLI


Windows PowerShell
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.
PS C:\Users\carmi> $ECR_URI = "${AWS_ACCOUNT_ID}.dkr.ecr.${AWS_REGION}.amazonaws.com/enterprise-microservice-app"
PS C:\Users\carmi> aws ecr get-login-password --region $AWS_REGION | docker login --username AWS --password-stdin "${AWS_ACCOUNT_ID}.dkr.ecr.${AWS_REGION}.amazonaws.com"
aws: [ERROR]: An error occurred (ParamValidation): argument --region: expected one argument
usage: aws [options] <command> <subcommand> [<subcommand> ...] [parameters]
To see help text, you can run:
aws help
aws <command> help
aws <command> <subcommand> help
password is empty
PS C:\Users\carmi> $AWS_REGION = "us-east-2"
PS C:\Users\carmi> $AWS_ACCOUNT_ID = (aws sts get-caller-identity --query Account --output text)
PS C:\Users\carmi> $ECR_URI = "${AWS_ACCOUNT_ID}.dkr.ecr.${AWS_REGION}.amazonaws.com/enterprise-microservice-app"
PS C:\Users\carmi> aws ecr get-login-password --region $AWS_REGION | docker login --username AWS --password-stdin "${AWS_ACCOUNT_ID}.dkr.ecr.${AWS_REGION}.amazonaws.com"
Login Succeeded
PS C:\Users\carmi>

Create Target Group
Target type IP
Protocol HTTP, Port 80


Elastic Container Registry - Pub
Target groups | EC2 | us-east-2
us-east-2.console.aws.amazon.com/ec2/home?region=...
aws
Search [Alt+S]
United State
carminecompa33 (586017285548)
carminecompa33
EC2 > Target groups
Target groups (1/1) Info | What's new?
Actions
Create target group
Filter target groups
1
Name
ARN
Port
Protocol
Target type
ecs-target-group
arn:aws:elasticloadbalancin...
80
HTTP
IP
Target group: ecs-target-group
Details
Targets
Monitoring
Health checks
Attributes
Tags
Details
arn:aws:elasticloadbalancing:us-east-2:586017285548:targetgroup/ecs-target-group/1cf0a443768f45c1
Target type
IP
Protocol : Port
HTTP: 80
Protocol version
HTTP1
VPC
vpc-07eee5771d15f8fc8
IP address type
IPv4
Load balancer
None associated

Create Security Group for my Application Load Balancer
Inbound rule: allow HTTP traffic on port 80 from anywhere

Elastic Container Registry - Pub

SecurityGroup | EC2 | us-east-2

us-east-2.console.aws.amazon.com/ec2/home?region=us-ea...

carminecompa33 (5860172...

carminecompa33

United States (O

EC2 > Security Groups > sg-0bf6e737ff18c21d3

Security group (sg-0bf6e737ff18c21d3 | alb-sg) was created successfully

Details

Details

Security group name	Security group ID	Description	VPC ID
alb-sg	sg-0bf6e737ff18c21d3	Allow inbound HTTP traffic to ALB	vpc-07eee5771d15f8fc8

Owner	Inbound rules count	Outbound rules count
586017285548	1 Permission entry	1 Permission entry

Inbound rules | Outbound rules | Sharing | VPC associations | Related

Inbound rules (1)

Manage tags | Edit inbound rules

Search

1

Created Load Balancer

HTTP:80

Forward to target group ecs-target-group

Elastic Container R | Load balancer det | Subnets | VPC | us-

us-east-2.console.aws.amazon.com/ec2/home?region=...

aws | Search [Alt+S] | United States (Ohio) | carminecompa33 (586017285548) carminecompa33

EC2 > Load balancers > ecs-app-alb

Successfully created load balancer: ecs-app-alb
It might take a few minutes for your load balancer to fully set up and route traffic. Targets will also take a few minutes to complete the registration process and pass initial health checks.

Introducing new Capacity Utilization metric for ALB
Monitor how much of your load balancer's available capacity was used to meet traffic demand. Visit the **Capacity** or **Monitoring** tabs to view utilization data.

ecs-app-alb

Actions

Details

Load balancer type
Application

Status
Provisioning

VPC
vpc-07eee5771d15f8fc8

Load balancer IP address type
IPv4

Scheme
Internet-facing

Hosted zone
Z3AADJGX6KTTL2

Availability Zones
subnet-098eeb16f98325e10
us-east-2a (use2-az1)
subnet-0492c83ab8e47e7ec
us-east-2b (use2-az2)

Date created
October 13, 2026, 06:50
(UTC+02:00)

Load balancer ARN
arn:aws:elasticloadbalancing:us-east-2:586017285548:loadbalancer/app/ecs-app-alb/90191774b1c6bb7e

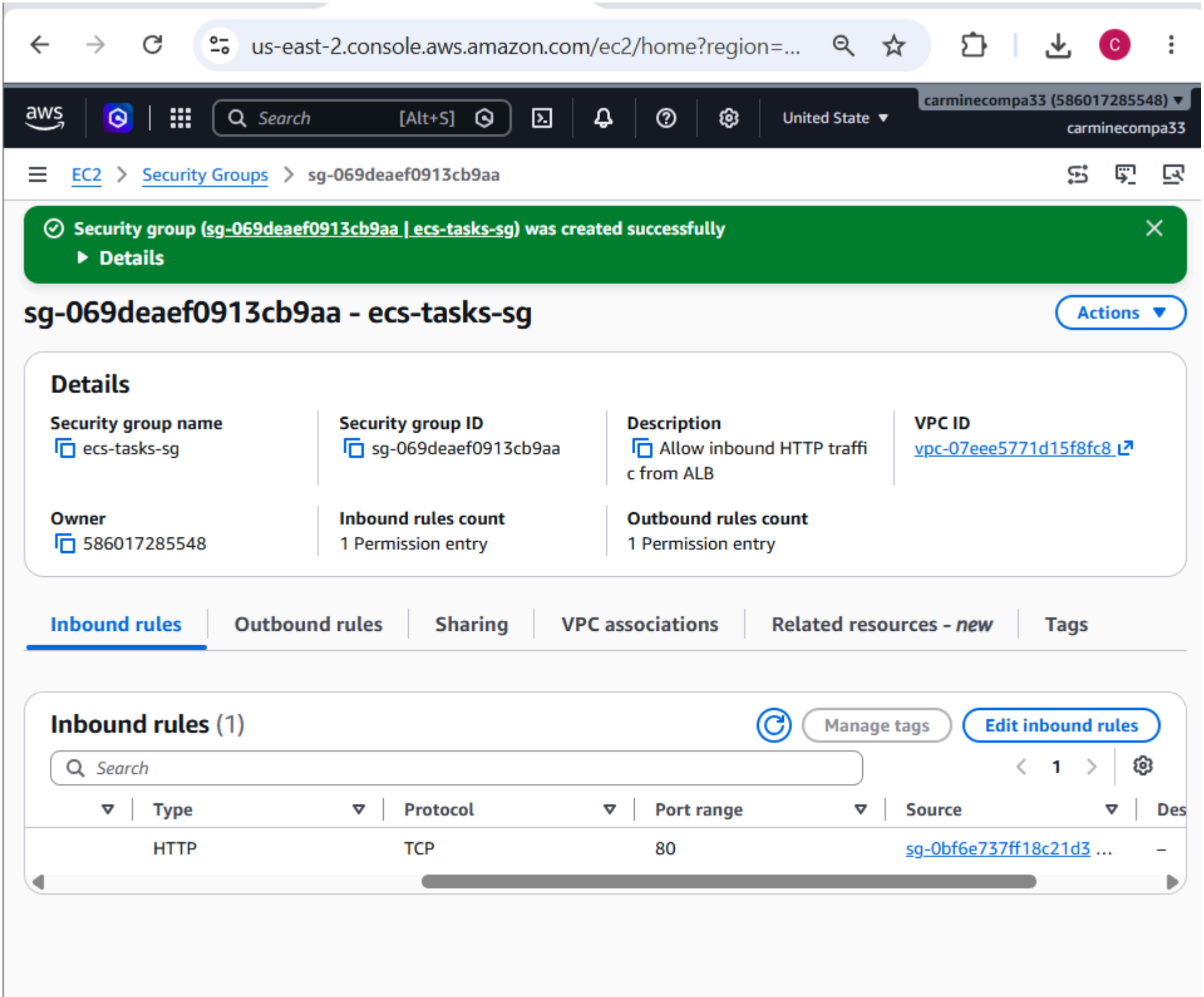
DNS name Info
ecs-app-alb-606278583.us-east-2.elb.amazonaws.com (A Record)

Listeners and rules | Network mapping | Resource map | Security | Monitoring | Integrations | Attributes

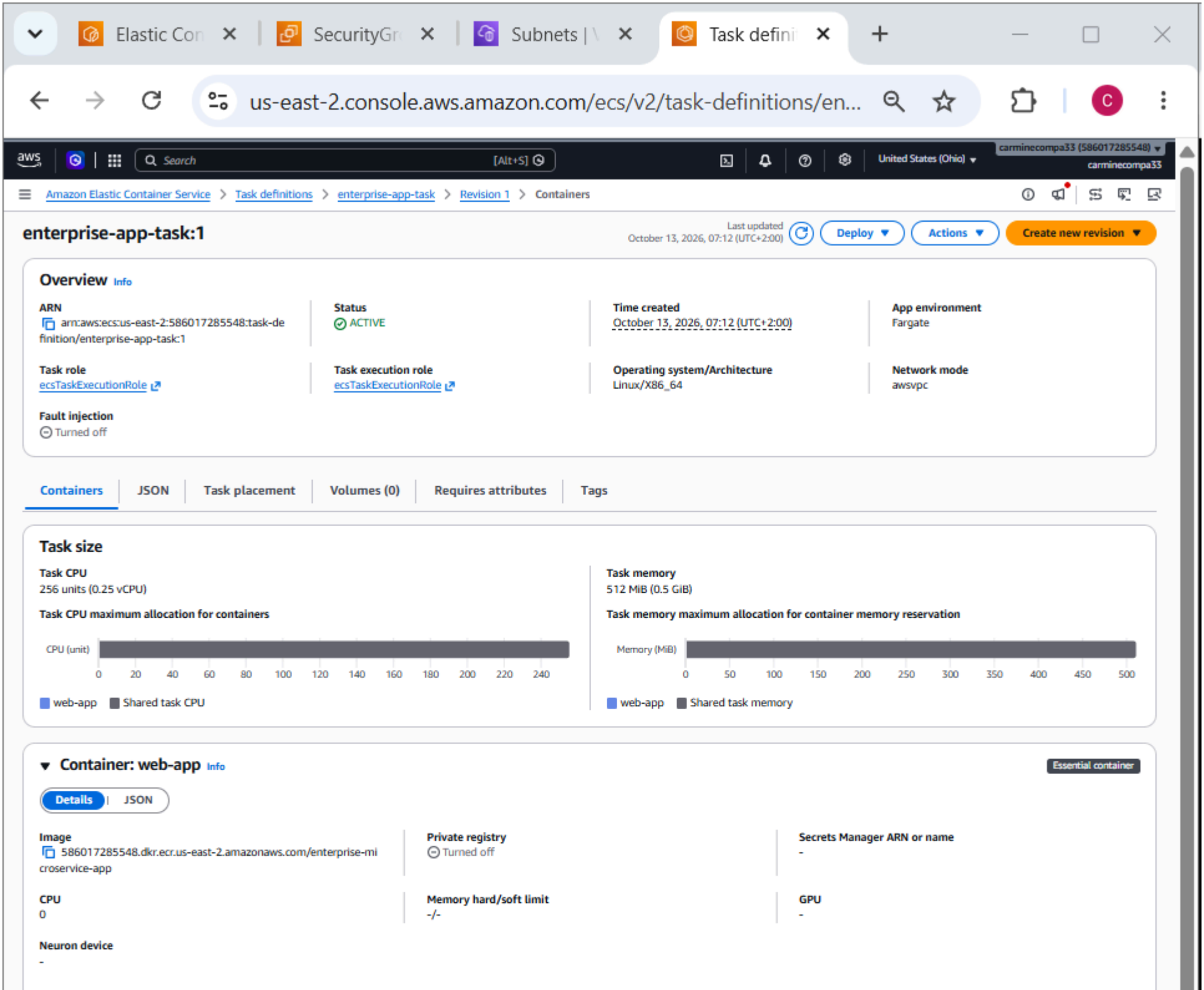
Listeners and rules (1) Info

Manage rules | Manage listener | Add listener

Create Security Group for EC2 tasks

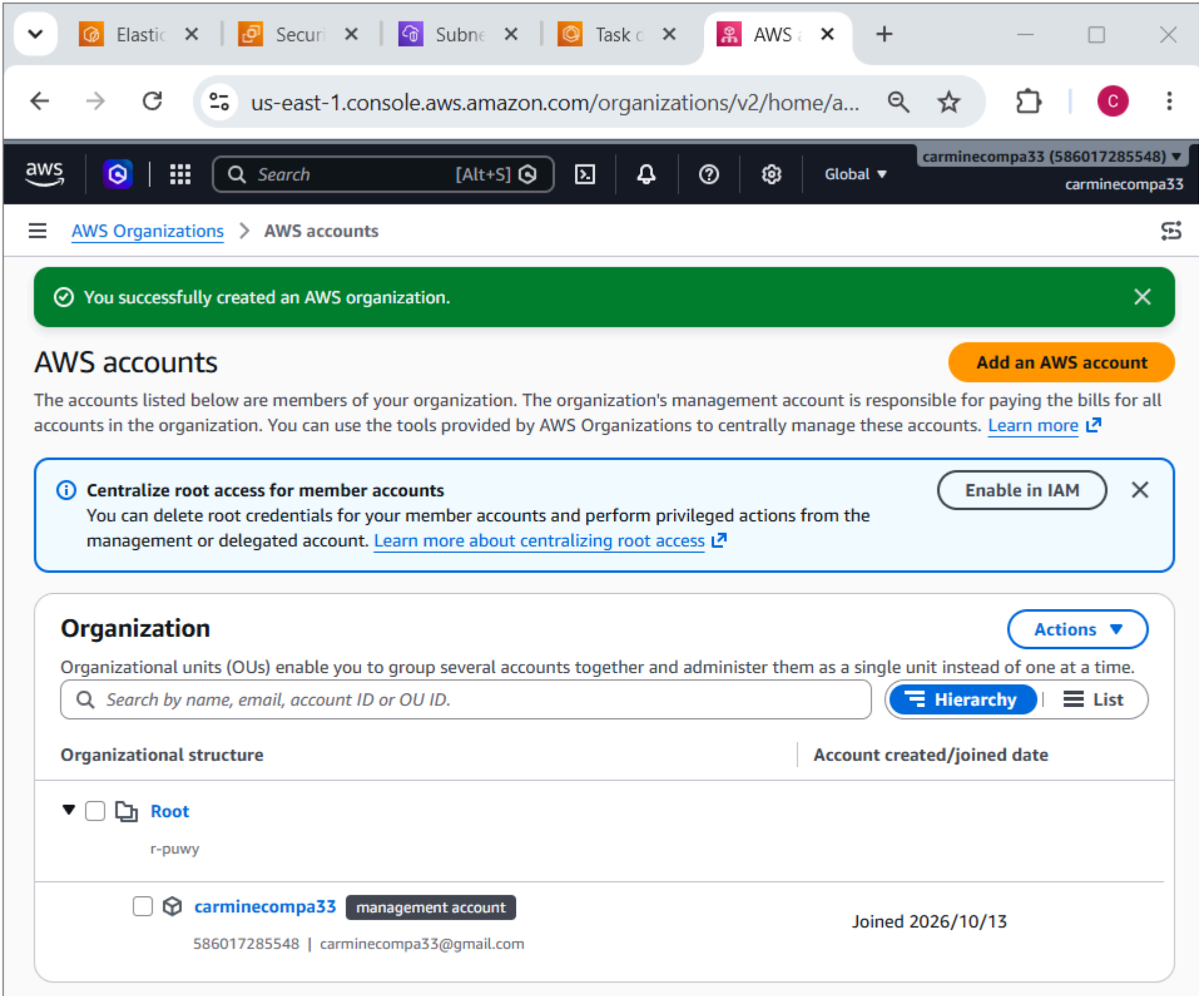

us-east-2.console.aws.amazon.com/ec2/home?region=...
aws
Search
[Alt+S]
United State
carminecompa33 (586017285548)
carminecompa33
EC2 > Security Groups > sg-069deaef0913cb9aa
Security group (sg-069deaef0913cb9aa | ecs-tasks-sg) was created successfully
Details
sg-069deaef0913cb9aa - ecs-tasks-sg
Actions
Details
Security group name
ecs-tasks-sg
Security group ID
sg-069deaef0913cb9aa
Description
Allow inbound HTTP traffi
c from ALB
VPC ID
vpc-07eee5771d15f8fc8
Owner
586017285548
Inbound rules count
1 Permission entry
Outbound rules count
1 Permission entry
Inbound rules
Outbound rules
Sharing
VPC associations
Related resources - new
Tags
Inbound rules (1)
Manage tags
Edit inbound rules
Search
1
Type
Protocol
Port range
Source
Des
HTTP
TCP
80
sg-0bf6e737ff18c21d3 ...
–

Created task definition with container from my ECR image URI


Elastic Con
SecurityGr
Subnets | \
Task defini
us-east-2.console.aws.amazon.com/ecs/v2/task-definitions/en...
Search
[Alt+S]
United States (Ohio)
carminecompa33 (586017285548)
carminecompa33
Amazon Elastic Container Service > Task definitions > enterprise-app-task > Revision 1 > Containers
enterprise-app-task:1
Last updated
October 13, 2026, 07:12 (UTC+2:00)
Deploy
Actions
Create new revision
Overview Info
ARN
arn:aws:ecs:us-east-2:586017285548:task-definition/enterprise-app-task:1
Status
ACTIVE
Time created
October 13, 2026, 07:12 (UTC+2:00)
App environment
Fargate
Task role
ecsTaskExecutionRole
Task execution role
ecsTaskExecutionRole
Operating system/Architecture
Linux/X86_64
Network mode
awsvpc
Fault injection
Turned off
Containers
JSON
Task placement
Volumes (0)
Requires attributes
Tags
Task size
Task CPU
256 units (0.25 vCPU)
Task memory
512 MiB (0.5 GiB)
Task CPU maximum allocation for containers
CPU (unit)
0 20 40 60 80 100 120 140 160 180 200 220 240
web-app
Shared task CPU
Task memory maximum allocation for container memory reservation
Memory (MiB)
0 50 100 150 200 250 300 350 400 450 500
web-app
Shared task memory
Container: web-app Info
Essential container
Details
JSON
Image
586017285548.dkr.ecr.us-east-2.amazonaws.com/enterprise-microservice-app
Private registry
Turned off
Secrets Manager ARN or name
-
CPU
0
Memory hard/soft limit
-/-
GPU
-
Neuron device
-

Created AWS Organization

Now my account is the Management Account


Elastic
Secur
Subne
Task c
AWS
us-east-1.console.aws.amazon.com/organizations/v2/home/a...
aws
Search
[Alt+S]
Global
carminecompa33 (586017285548)
carminecompa33
AWS Organizations
AWS accounts
You successfully created an AWS organization.
AWS accounts
Add an AWS account
The accounts listed below are members of your organization. The organization's management account is responsible for paying the bills for all accounts in the organization. You can use the tools provided by AWS Organizations to centrally manage these accounts. Learn more
Centralize root access for member accounts
You can delete root credentials for your member accounts and perform privileged actions from the management or delegated account. Learn more about centralizing root access
Enable in IAM
Organization
Actions
Organizational units (OUs) enable you to group several accounts together and administer them as a single unit instead of one at a time.
Search by name, email, account ID or OU ID.
Hierarchy
List
Organizational structure
Account created/joined date
Root
r-puwy
carminecompa33
management account
586017285548 | carminecompa33@gmail.com
Joined 2026/10/13

Created SCP policy

This) acts as a preventive guardrail that restricts resource creation outside of designated AWS regions

Successfully created the policy named 'DenyUnapprovedRegionsAndUnencryptedEBS'.

DenyUnapprovedRegionsAndUnencryptedEBS

Delete | Edit policy

Policy details

Name
DenyUnapprovedRegionsAndUnencryptedEBS

ARN
arn:aws:organizations::586017285548:policy/o-kz7dj2ai57/service_control_policy/p-p46c5w4v

Policy type
Service control policy (customer managed)

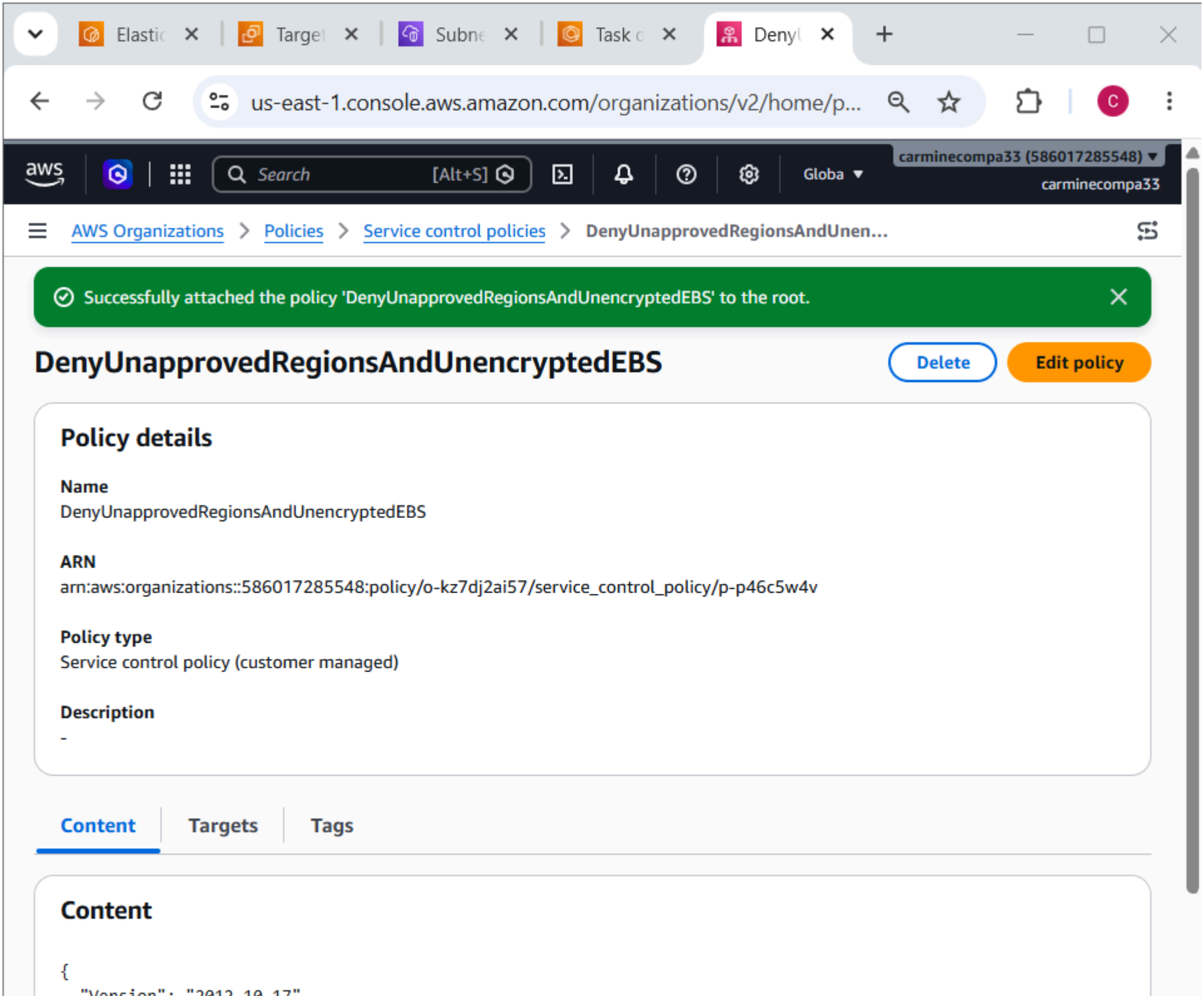
Description
-

Content | Targets | Tags

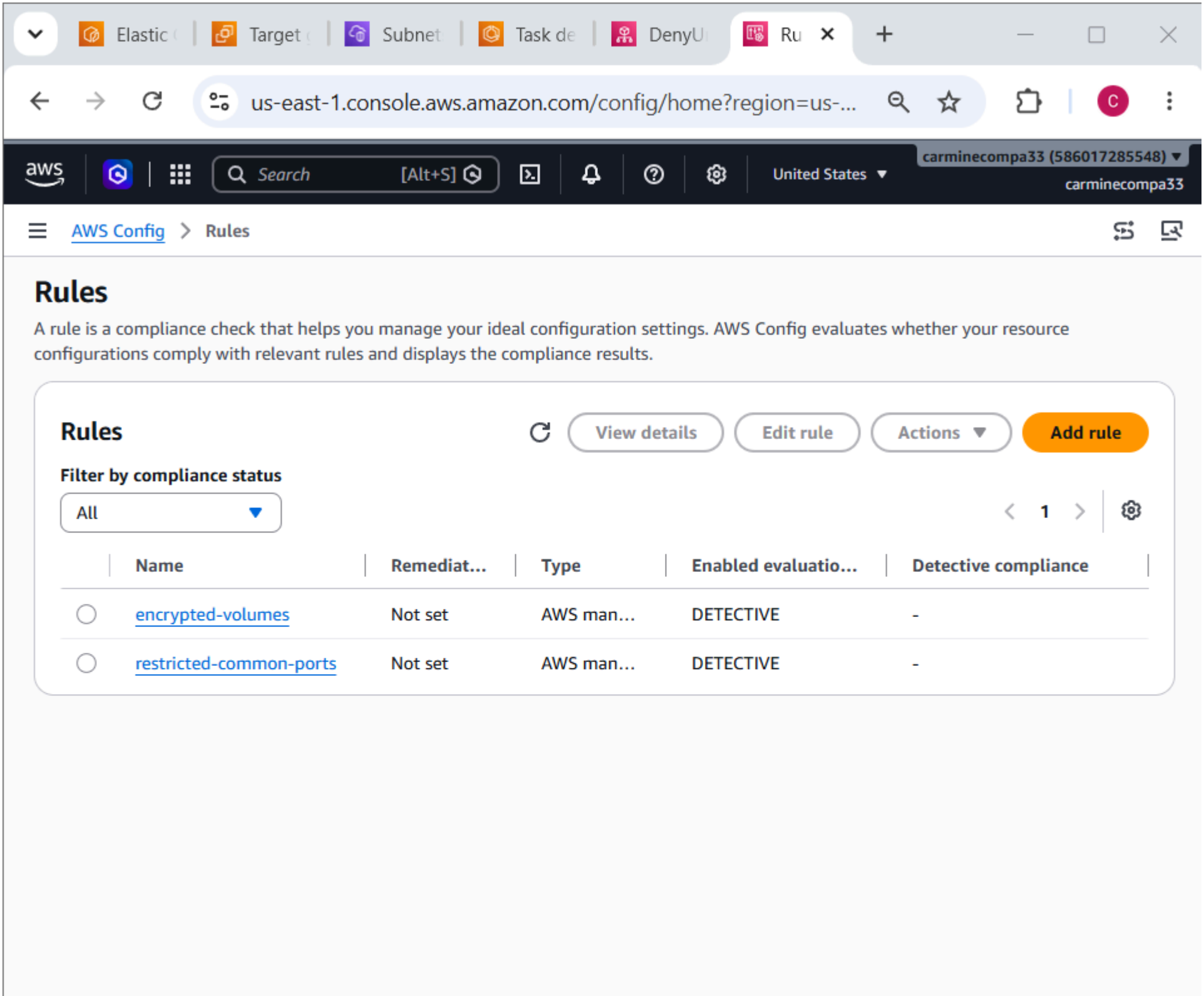
Content

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "DenyAllOutsideApprovedRegions",
      "Effect": "Deny",
      "NotAction": [
        "iam:*",
        "organizations:*",
        "route53:*",
        "cloudfront:*",
        "sts:*",
        "support:*",
        "securityhub:*",
        "config:*"
      ],
      "Resource": "*",
      "Condition": {
        "StringNotEquals": {
          "aws:RequestedRegion": [
            "us-east-2",
            "us-east-1"
          ]
        }
      }
    }
  ]
}
```

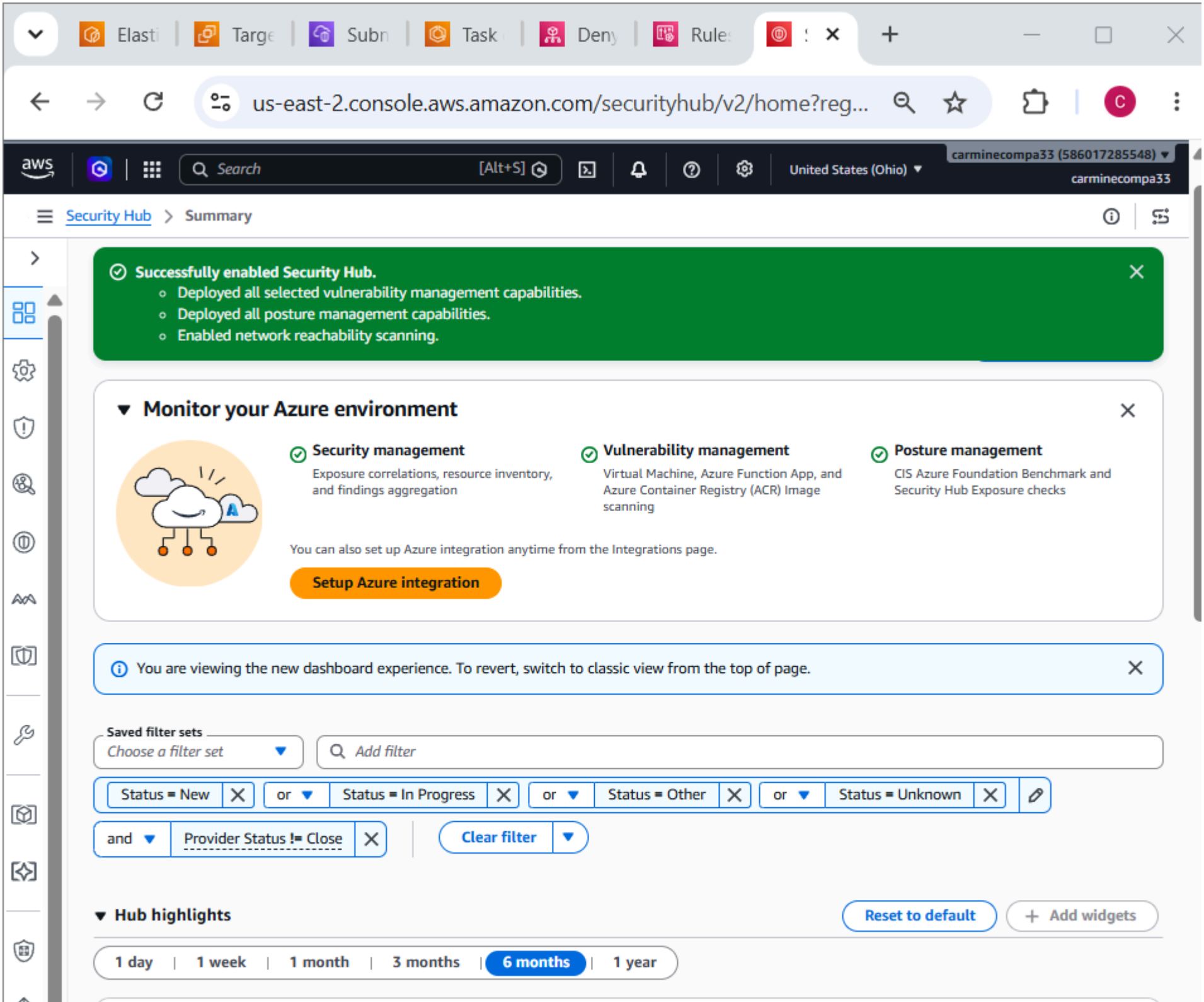
I attached the SCP policy to the AWS Organization


Elastic
Target
Subne
Task c
Deny
us-east-1.console.aws.amazon.com/organizations/v2/home/p...
aws
Search
[Alt+S]
Globa
carminecompa33 (586017285548)
carminecompa33
AWS Organizations
Policies
Service control policies
DenyUnapprovedRegionsAndUnen...
Successfully attached the policy 'DenyUnapprovedRegionsAndUnencryptedEBS' to the root.
DenyUnapprovedRegionsAndUnencryptedEBS
Delete
Edit policy
Policy details
Name
DenyUnapprovedRegionsAndUnencryptedEBS
ARN
arn:aws:organizations::586017285548:policy/o-kz7dj2ai57/service_control_policy/p-p46c5w4v
Policy type
Service control policy (customer managed)
Description
-
Content
Targets
Tags
Content
{

Added rules to the AWS Config


Elastic
Target
Subnet
Task de
DenyU
Ru
us-east-1.console.aws.amazon.com/config/home?region=us-...
aws
Search
[Alt+S]
United States
carminecompa33 (586017285548)
carminecompa33
AWS Config
Rules
Rules
A rule is a compliance check that helps you manage your ideal configuration settings. AWS Config evaluates whether your resource configurations comply with relevant rules and displays the compliance results.
Rules
View details
Edit rule
Actions
Add rule
Filter by compliance status
All
1
Name
Remediat...
Type
Enabled evaluatio...
Detective compliance
encrypted-volumes
Not set
AWS man...
DETECTIVE
-
restricted-common-ports
Not set
AWS man...
DETECTIVE
-

Enabled Security Hub


us-east-2.console.aws.amazon.com/securityhub/v2/home?reg...
Search
[Alt+S]
United States (Ohio)
carminecompa33 (586017285548)
carminecompa33
Security Hub > Summary
Successfully enabled Security Hub.
Deployed all selected vulnerability management capabilities.
Deployed all posture management capabilities.
Enabled network reachability scanning.
Monitor your Azure environment
Security management
Exposure correlations, resource inventory, and findings aggregation
Vulnerability management
Virtual Machine, Azure Function App, and Azure Container Registry (ACR) Image scanning
Posture management
CIS Azure Foundation Benchmark and Security Hub Exposure checks
You can also set up Azure integration anytime from the Integrations page.
Setup Azure integration
You are viewing the new dashboard experience. To revert, switch to classic view from the top of page.
Saved filter sets
Choose a filter set
Add filter
Status = New
or
Status = In Progress
or
Status = Other
or
Status = Unknown
and
Provider Status != Close
Clear filter
Hub highlights
Reset to default
Add widgets
1 day
1 week
1 month
3 months
6 months
1 year

Build and tag Docker Image with the ECR URI

Windows PowerShell

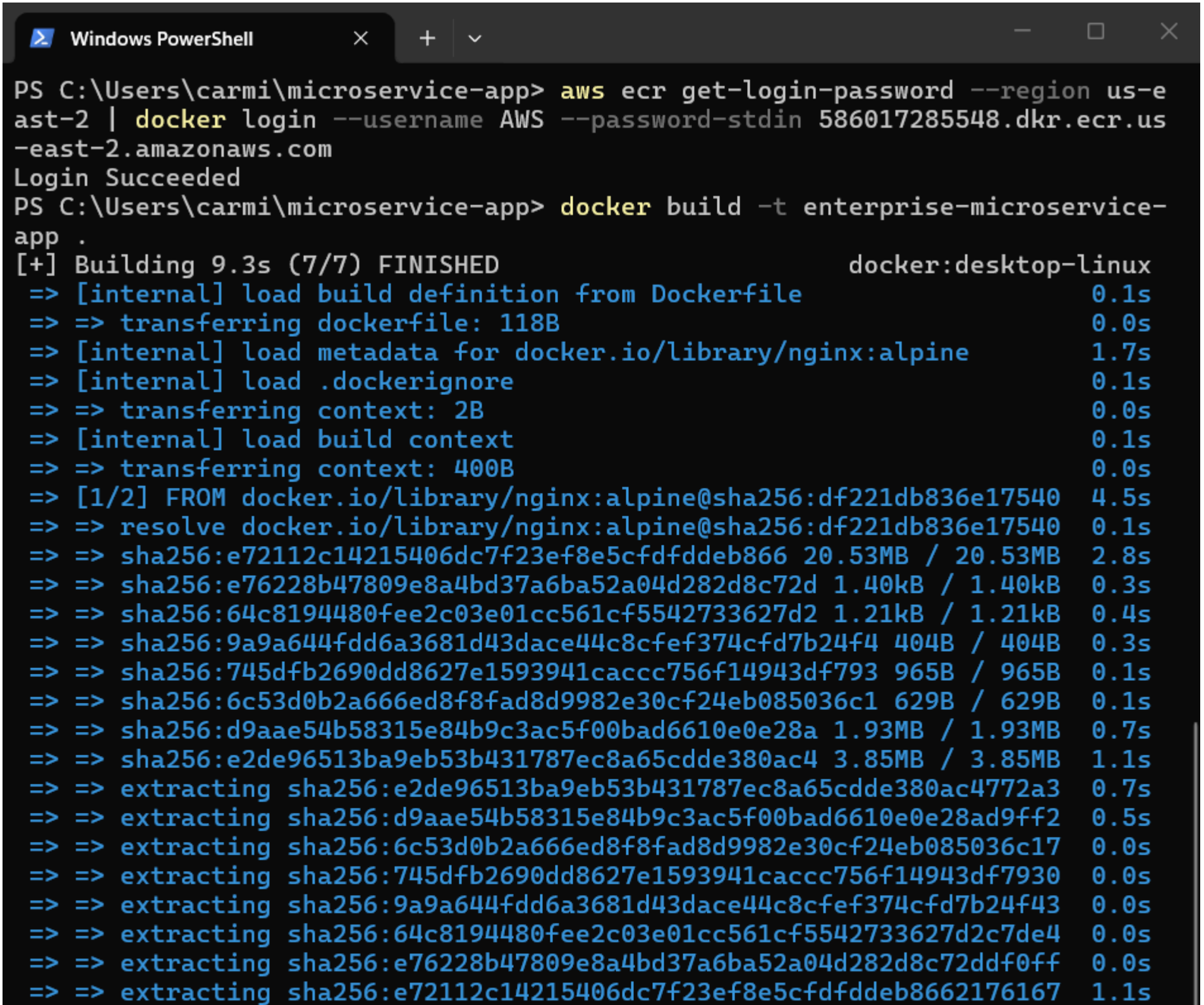
```
PS C:\Users\carmi> aws ecr describe-images --repository-name enterprise-micr
oservice-app --region $AWS_REGION
{
    "imageDetails": []
}

PS C:\Users\carmi> mkdir $HOME\microservice-app;


    Directory: C:\Users\carmi


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
d-----        10/14/2026   7:26 AM                microservice-app


PS C:\Users\carmi> cd $HOME\microservice-app
PS C:\Users\carmi\microservice-app> @'
>> <!DOCTYPE html>
>> <html>
>> <head><title>Fargate Microservice</title></head>
>> <body style="font-family: Arial; text-align: center; padding-top: 50px; b
ackground-color: #f4f6f8;">
>>   <h1 style="color: #232f3e;">Enterprise Microservice App Running on Farg
ate</h1>
>>   <p style="color: #ff9900; font-size: 1.2em;">Status: Healthy & Deployed
 via ALB</p>
>> </body>
>> </html>
>> '@ | Out-File -FilePath index.html -Encoding utf8
PS C:\Users\carmi\microservice-app>
```


Windows PowerShell
PS C:\Users\carmi\microservice-app> aws ecr get-login-password --region us-east-2 | docker login --username AWS --password-stdin 586017285548.dkr.ecr.us-east-2.amazonaws.com
Login Succeeded
PS C:\Users\carmi\microservice-app> docker build -t enterprise-microservice-app .
[+] Building 9.3s (7/7) FINISHED docker:desktop-linux
=> [internal] load build definition from Dockerfile 0.1s
=> => transferring dockerfile: 118B 0.0s
=> [internal] load metadata for docker.io/library/nginx:alpine 1.7s
=> [internal] load .dockerignore 0.1s
=> => transferring context: 2B 0.0s
=> [internal] load build context 0.1s
=> => transferring context: 400B 0.0s
=> [1/2] FROM docker.io/library/nginx:alpine@sha256:df221db836e17540 4.5s
=> => resolve docker.io/library/nginx:alpine@sha256:df221db836e17540 0.1s
=> => sha256:e72112c14215406dc7f23ef8e5cfdfddeb866 20.53MB / 20.53MB 2.8s
=> => sha256:e76228b47809e8a4bd37a6ba52a04d282d8c72d 1.40kB / 1.40kB 0.3s
=> => sha256:64c8194480fee2c03e01cc561cf5542733627d2 1.21kB / 1.21kB 0.4s
=> => sha256:9a9a644fdd6a3681d43dace44c8cfef374cfd7b24f4 404B / 404B 0.3s
=> => sha256:745dfb2690dd8627e1593941caccc756f14943df793 965B / 965B 0.1s
=> => sha256:6c53d0b2a666ed8f8fad8d9982e30cf24eb085036c1 629B / 629B 0.1s
=> => sha256:d9aae54b58315e84b9c3ac5f00bad6610e0e28a 1.93MB / 1.93MB 0.7s
=> => sha256:e2de96513ba9eb53b431787ec8a65cdde380ac4 3.85MB / 3.85MB 1.1s
=> => extracting sha256:e2de96513ba9eb53b431787ec8a65cdde380ac4772a3 0.7s
=> => extracting sha256:d9aae54b58315e84b9c3ac5f00bad6610e0e28ad9ff2 0.5s
=> => extracting sha256:6c53d0b2a666ed8f8fad8d9982e30cf24eb085036c17 0.0s
=> => extracting sha256:745dfb2690dd8627e1593941caccc756f14943df7930 0.0s
=> => extracting sha256:9a9a644fdd6a3681d43dace44c8cfef374cfd7b24f43 0.0s
=> => extracting sha256:64c8194480fee2c03e01cc561cf5542733627d2c7de4 0.0s
=> => extracting sha256:e76228b47809e8a4bd37a6ba52a04d282d8c72ddf0ff 0.0s
=> => extracting sha256:e72112c14215406dc7f23ef8e5cfdfddeb8662176167 1.1s

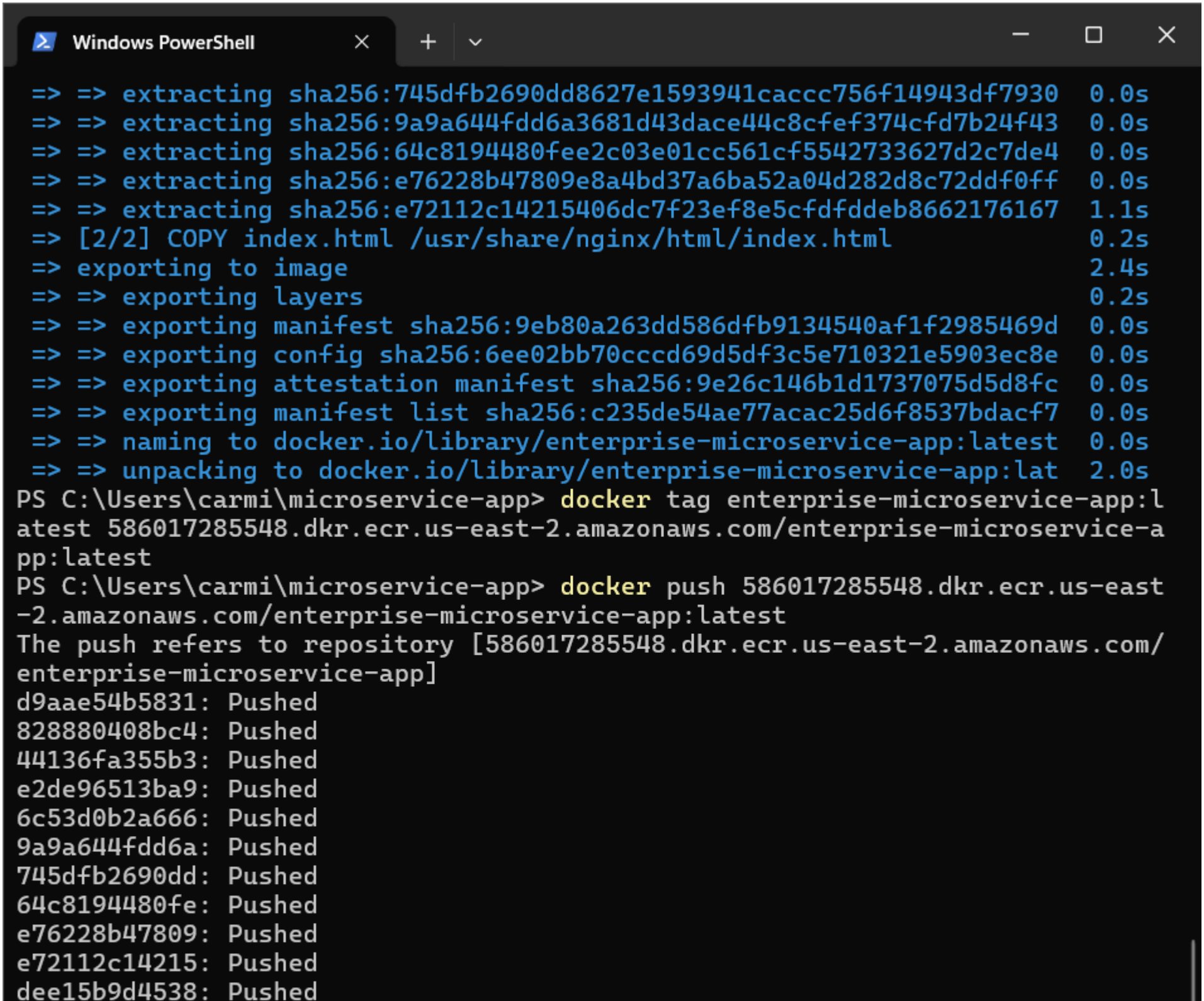
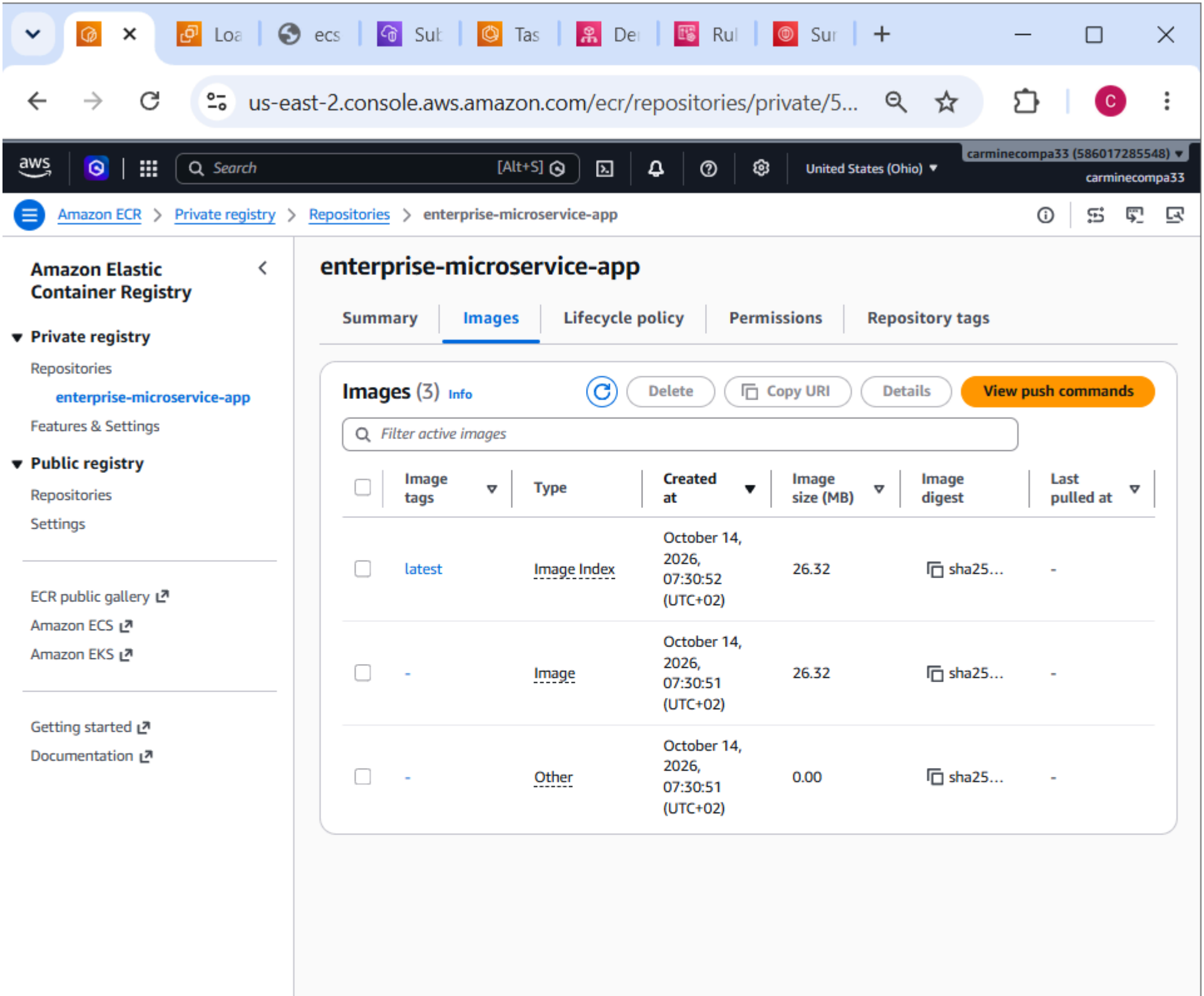
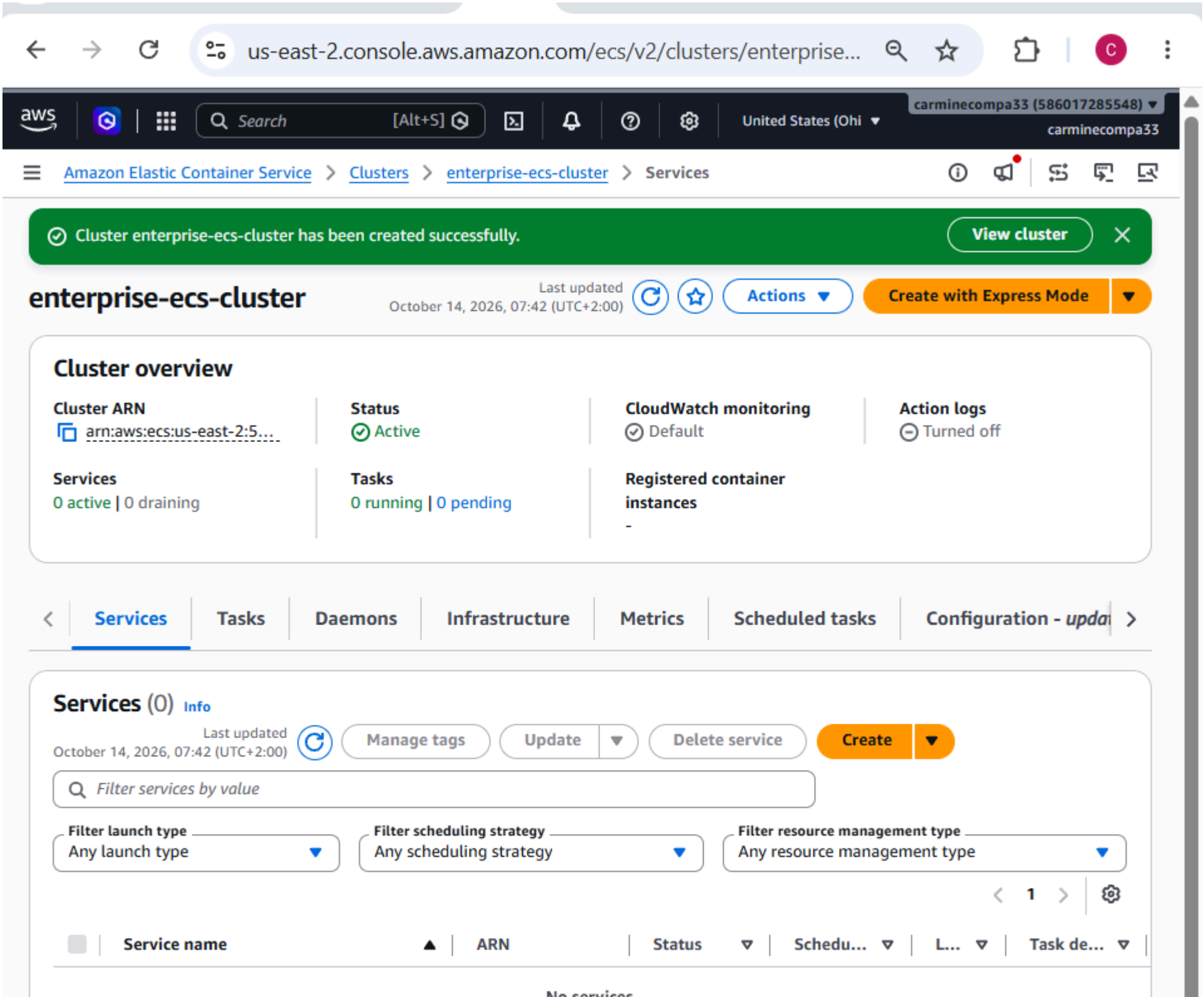

Windows PowerShell
=> => extracting sha256:745dfb2690dd8627e1593941caccc756f14943df7930 0.0s
=> => extracting sha256:9a9a644fdd6a3681d43dace44c8cfef374cfd7b24f43 0.0s
=> => extracting sha256:64c8194480fee2c03e01cc561cf5542733627d2c7de4 0.0s
=> => extracting sha256:e76228b47809e8a4bd37a6ba52a04d282d8c72ddf0ff 0.0s
=> => extracting sha256:e72112c14215406dc7f23ef8e5cfdfddeb8662176167 1.1s
=> [2/2] COPY index.html /usr/share/nginx/html/index.html 0.2s
=> exporting to image 2.4s
=> => exporting layers 0.2s
=> => exporting manifest sha256:9eb80a263dd586dfb9134540af1f2985469d 0.0s
=> => exporting config sha256:6ee02bb70cccd69d5df3c5e710321e5903ec8e 0.0s
=> => exporting attestation manifest sha256:9e26c146b1d1737075d5d8fc 0.0s
=> => exporting manifest list sha256:c235de54ae77acac25d6f8537bdacf7 0.0s
=> => naming to docker.io/library/enterprise-microservice-app:latest 0.0s
=> => unpacking to docker.io/library/enterprise-microservice-app:lat 2.0s
PS C:\Users\carmi\microservice-app> docker tag enterprise-microservice-app:latest 586017285548.dkr.ecr.us-east-2.amazonaws.com/enterprise-microservice-app:latest
PS C:\Users\carmi\microservice-app> docker push 586017285548.dkr.ecr.us-east-2.amazonaws.com/enterprise-microservice-app:latest
The push refers to repository [586017285548.dkr.ecr.us-east-2.amazonaws.com/enterprise-microservice-app]
d9aae54b5831: Pushed
828880408bc4: Pushed
44136fa355b3: Pushed
e2de96513ba9: Pushed
6c53d0b2a666: Pushed
9a9a644fdd6a: Pushed
745dfb2690dd: Pushed
64c8194480fe: Pushed
e76228b47809: Pushed
e72112c14215: Pushed
dee15b9d4538: Pushed

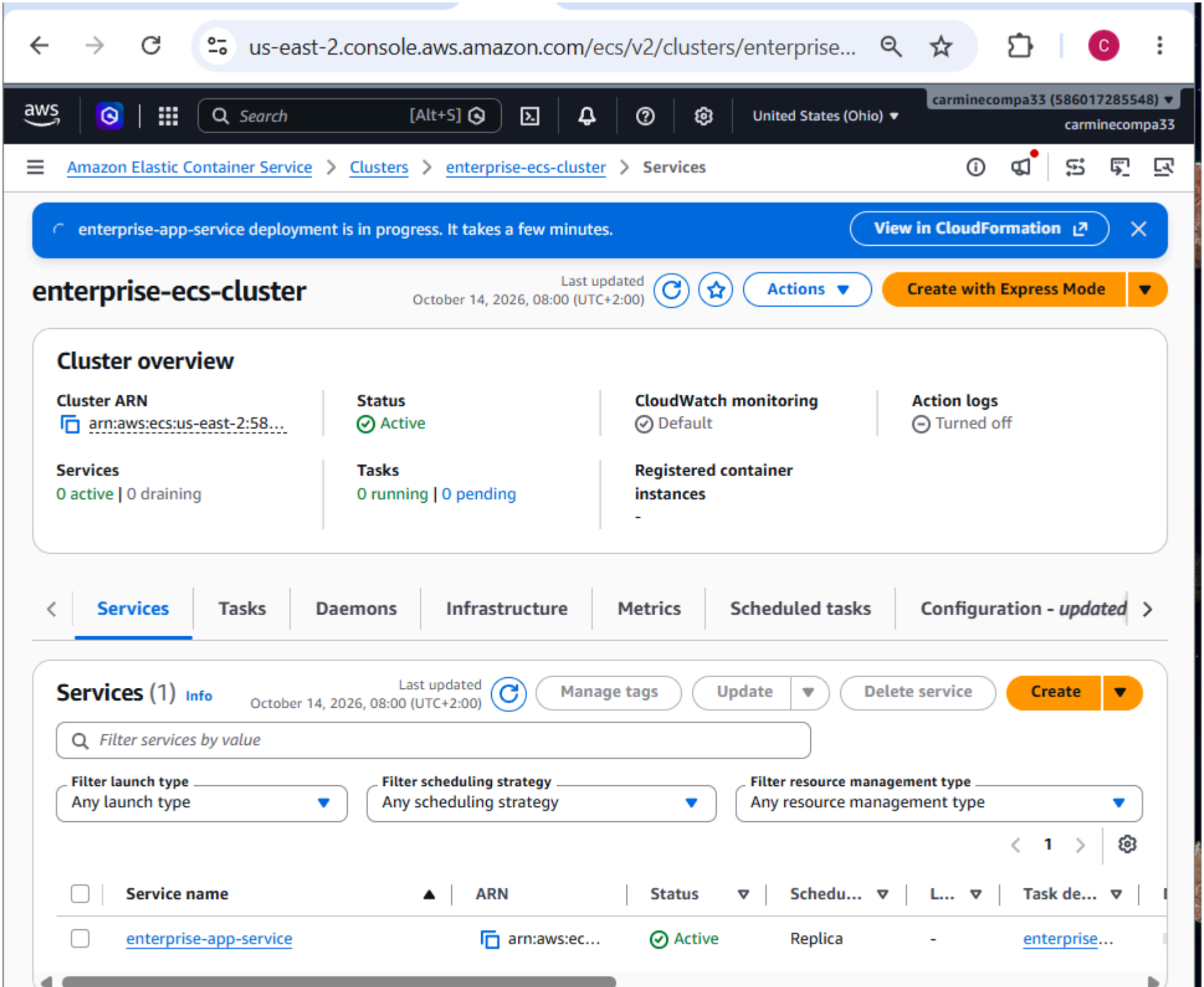
Image successfully pushed to the ECR in AWS

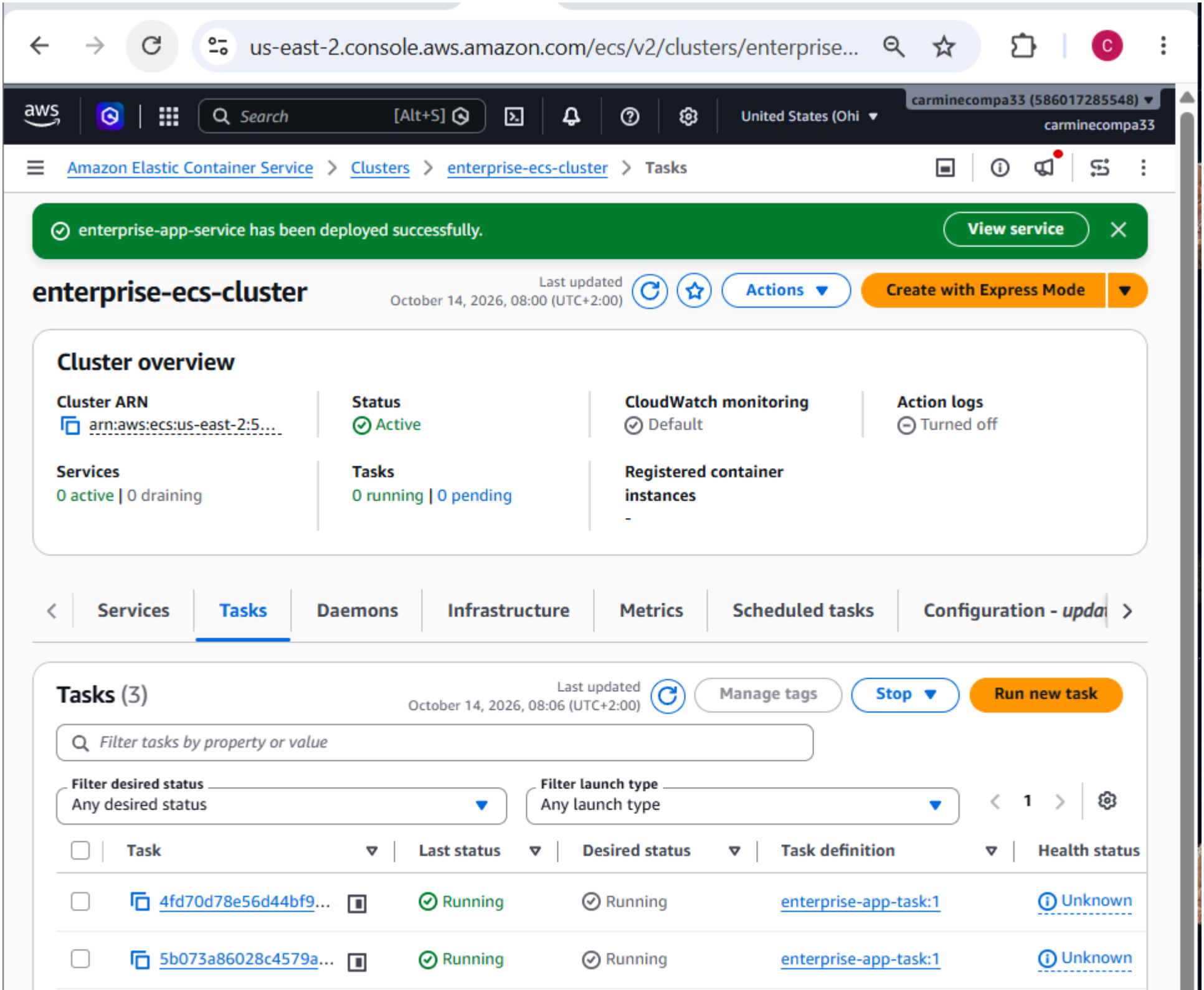

us-east-2.console.aws.amazon.com/ecr/repositories/private/5...
Search [Alt+S]
United States (Ohio)
carminecompa33 (586017285548)
carminecompa33
Amazon ECR > Private registry > Repositories > enterprise-microservice-app
Amazon Elastic Container Registry
Private registry
Repositories
enterprise-microservice-app
Features & Settings
Public registry
Repositories
Settings
ECR public gallery
Amazon ECS
Amazon EKS
Getting started
Documentation
enterprise-microservice-app
Summary
Images
Lifecycle policy
Permissions
Repository tags
Images (3) Info
Delete
Copy URI
Details
View push commands
Filter active images
Image tags
Type
Created at
Image size (MB)
Image digest
Last pulled at
latest
Image Index
October 14, 2026, 07:30:52 (UTC+02)
26.32
sha25...
-
-
Image
October 14, 2026, 07:30:51 (UTC+02)
26.32
sha25...
-
-
Other
October 14, 2026, 07:30:51 (UTC+02)
0.00
sha25...
-

Created ECS fargate cluster

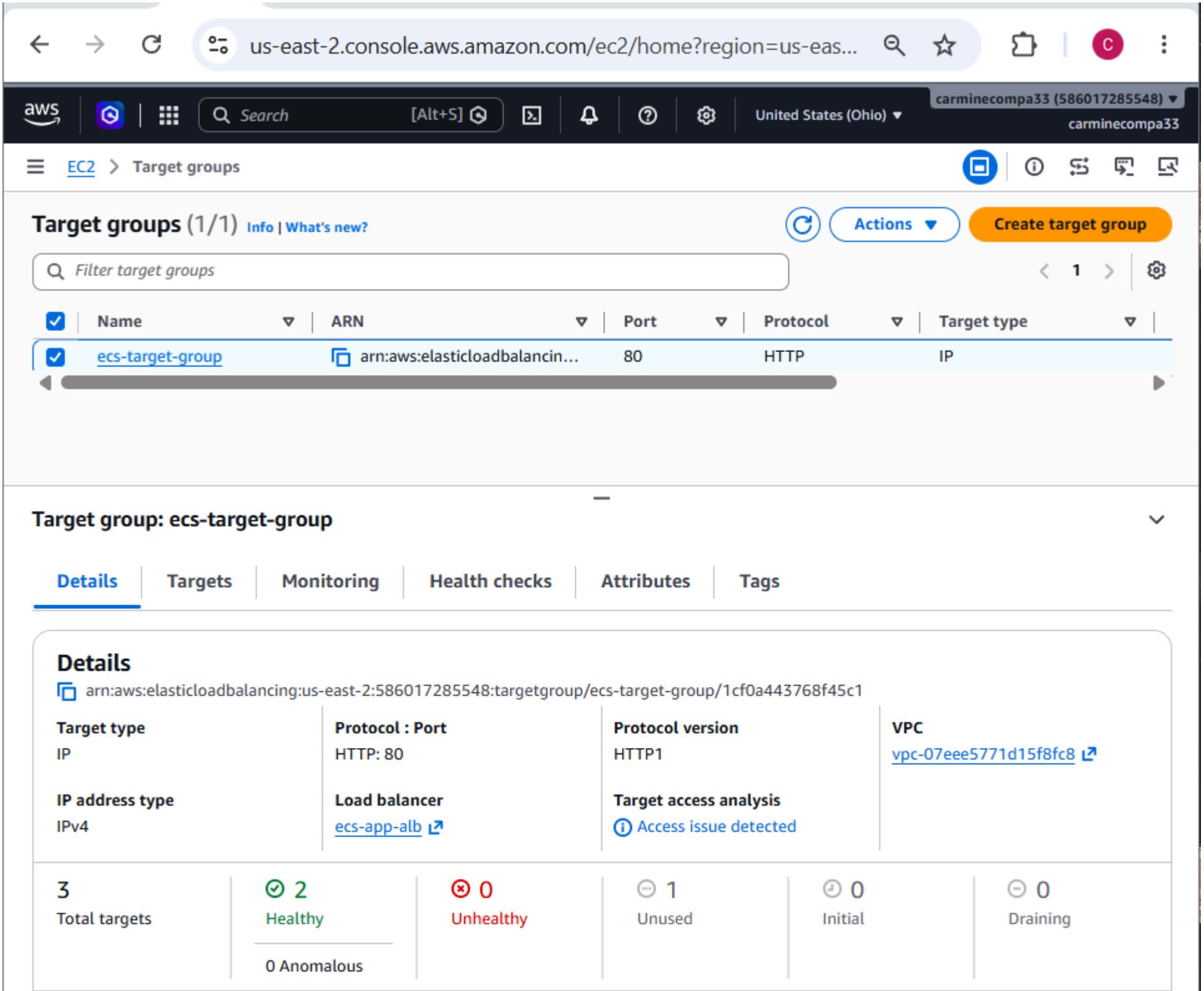

us-east-2.console.aws.amazon.com/ecs/v2/clusters/enterprise...
aws
Search
[Alt+S]
United States (Ohi
carminecompa33 (586017285548)
carminecompa33
Amazon Elastic Container Service
Clusters
enterprise-ecs-cluster
Services
Cluster enterprise-ecs-cluster has been created successfully.
View cluster
enterprise-ecs-cluster
Last updated
October 14, 2026, 07:42 (UTC+2:00)
Actions
Create with Express Mode
Cluster overview
Cluster ARN
arn:aws:ecs:us-east-2:5...
Status
Active
CloudWatch monitoring
Default
Action logs
Turned off
Services
0 active | 0 draining
Tasks
0 running | 0 pending
Registered container instances
-
Services
Tasks
Daemons
Infrastructure
Metrics
Scheduled tasks
Configuration - upda
Services (0) Info
Last updated
October 14, 2026, 07:42 (UTC+2:00)
Manage tags
Update
Delete service
Create
Filter services by value
Filter launch type
Any launch type
Filter scheduling strategy
Any scheduling strategy
Filter resource management type
Any resource management type
1
Service name
ARN
Status
Schedu...
L...
Task de...
No services

Created ECS cluster

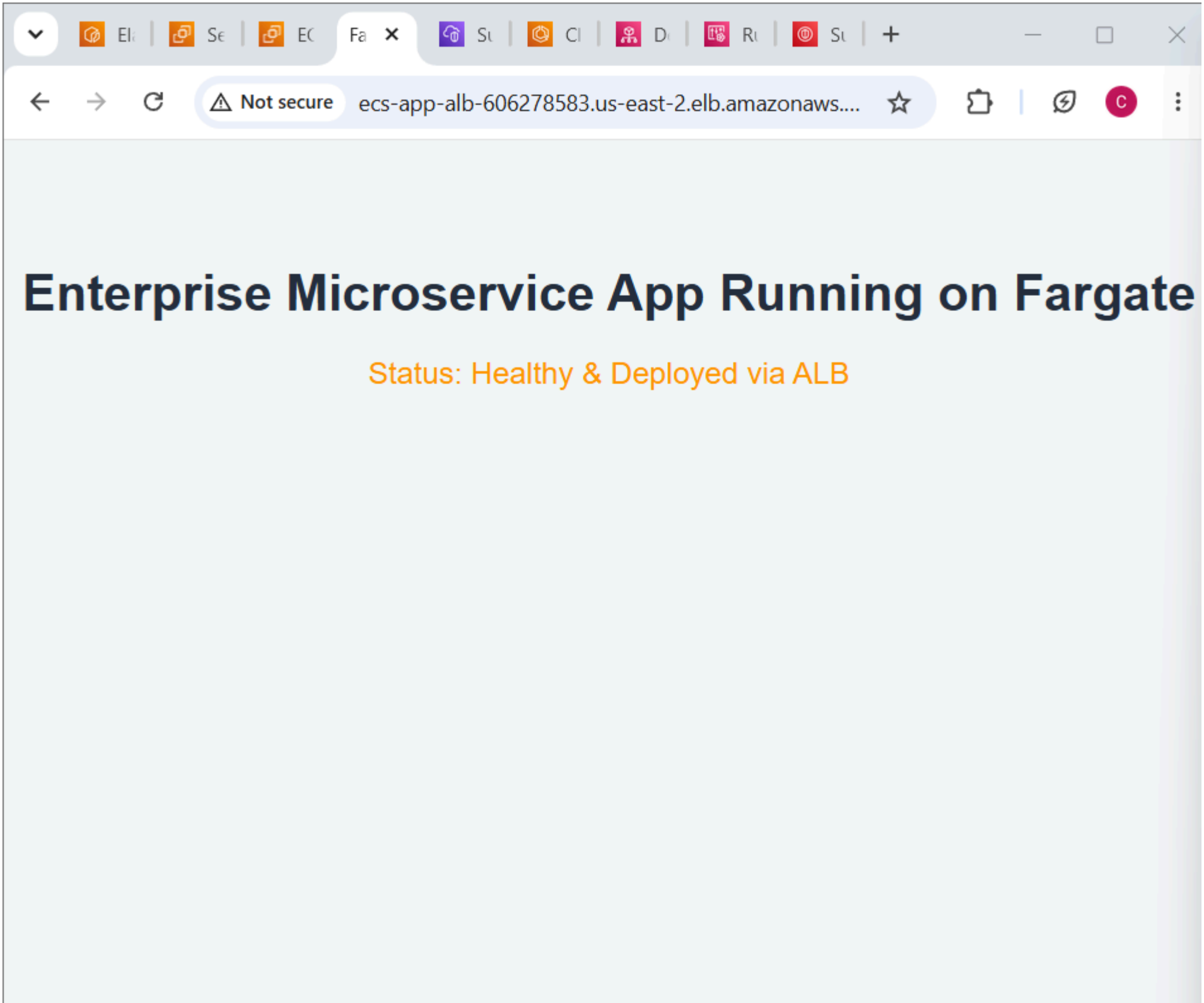

us-east-2.console.aws.amazon.com/ecs/v2/clusters/enterprise...
aws
Search
[Alt+S]
United States (Ohio)
carminecompa33 (586017285548)
carminecompa33
Amazon Elastic Container Service
Clusters
enterprise-ecs-cluster
Services
enterprise-app-service deployment is in progress. It takes a few minutes.
View in CloudFormation
enterprise-ecs-cluster
Last updated
October 14, 2026, 08:00 (UTC+2:00)
Actions
Create with Express Mode
Cluster overview
Cluster ARN
arn:aws:ecs:us-east-2:58...
Status
Active
CloudWatch monitoring
Default
Action logs
Turned off
Services
0 active | 0 draining
Tasks
0 running | 0 pending
Registered container instances
-
Services
Tasks
Daemons
Infrastructure
Metrics
Scheduled tasks
Configuration - updated
Services (1) Info
Last updated
October 14, 2026, 08:00 (UTC+2:00)
Manage tags
Update
Delete service
Create
Filter services by value
Filter launch type
Any launch type
Filter scheduling strategy
Any scheduling strategy
Filter resource management type
Any resource management type
1
Service name
ARN
Status
Schedu...
L...
Task de...
enterprise-app-service
arn:aws:ec...
Active
Replica
-
enterprise...


us-east-2.console.aws.amazon.com/ecs/v2/clusters/enterprise...
aws
Search
[Alt+S]
United States (Ohi
carminecompa33 (586017285548)
carminecompa33
Amazon Elastic Container Service
Clusters
enterprise-ecs-cluster
Tasks
enterprise-app-service has been deployed successfully.
View service
enterprise-ecs-cluster
Last updated
October 14, 2026, 08:00 (UTC+2:00)
Actions
Create with Express Mode
Cluster overview
Cluster ARN
arn:aws:ecs:us-east-2:5...
Status
Active
CloudWatch monitoring
Default
Action logs
Turned off
Services
0 active | 0 draining
Tasks
0 running | 0 pending
Registered container instances
-
Services
Tasks
Daemons
Infrastructure
Metrics
Scheduled tasks
Configuration - upda
Tasks (3)
Last updated
October 14, 2026, 08:06 (UTC+2:00)
Manage tags
Stop
Run new task
Filter tasks by property or value
Filter desired status
Any desired status
Filter launch type
Any launch type
1
Task
Last status
Desired status
Task definition
Health status
4fd70d78e56d44bf9...
Running
Running
enterprise-app-task:1
Unknown
5b073a86028c4579a...
Running
Running
enterprise-app-task:1
Unknown

Target Group is now healthy with 2 targets

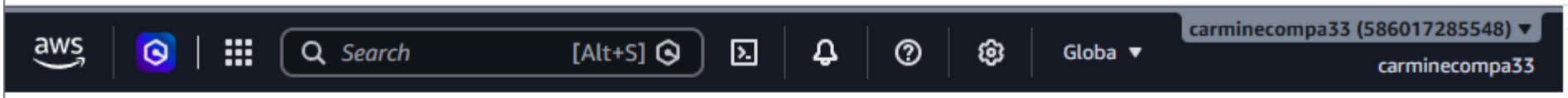

us-east-2.console.aws.amazon.com/ec2/home?region=us-eas...
aws
Search
[Alt+S]
United States (Ohio)
carminecompa33 (586017285548)
carminecompa33
EC2 > Target groups
Target groups (1/1) Info | What's new?
Actions
Create target group
Filter target groups
1
Name
ARN
Port
Protocol
Target type
ecs-target-group
arn:aws:elasticloadbalancin...
80
HTTP
IP
Target group: ecs-target-group
Details
Targets
Monitoring
Health checks
Attributes
Tags
Details
arn:aws:elasticloadbalancing:us-east-2:586017285548:targetgroup/ecs-target-group/1cf0a443768f45c1
Target type
IP
Protocol : Port
HTTP: 80
Protocol version
HTTP1
VPC
vpc-07eee5771d15f8fc8
IP address type
IPv4
Load balancer
ecs-app-alb
Target access analysis
Access issue detected
3
Total targets
2
Healthy
0 Anomalous
0
Unhealthy
1
Unused
0
Initial
0
Draining

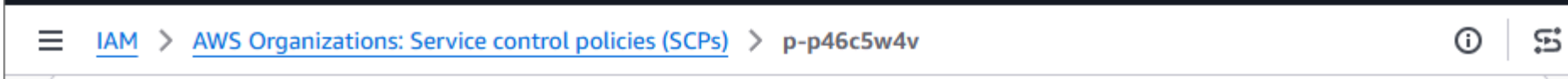
Application Load Balancer DNS name works fine


Not secure
ecs-app-alb-606278583.us-east-2.elb.amazonaws....
Enterprise Microservice App Running on Fargate
Status: Healthy & Deployed via ALB

Now I have to check if my SCP policy is actually working for Member Accounts
Below the SCP policy DenyUnapprovedRegionsAndUnencryptedEBS







Service access report (0)

For each service, view which account last accessed the service and when. This data can help you decide whether to edit the SCP in the AWS Organizations console to deny access to services you don't need. Data is available for services that are allowed by directly attached SCPs only. Recent activity usually appears within 4 hours. Data is stored for a maximum of 400 days. Learn more

Search

1

Service	Last accessed	Last accessed by account

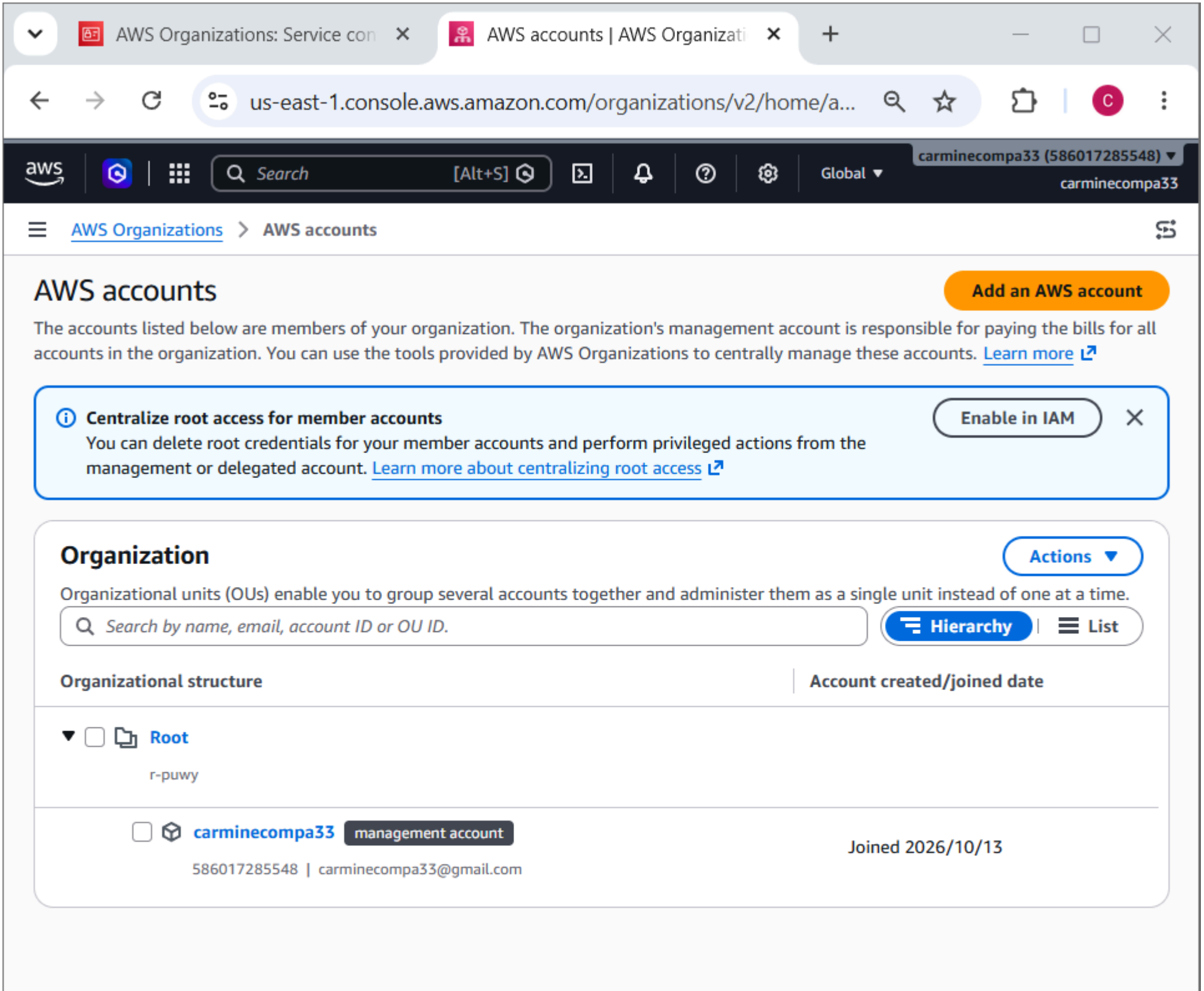
No accessible services to display. Ensure the attached control policy grants permissions to at least one service. Learn more

Policy JSON

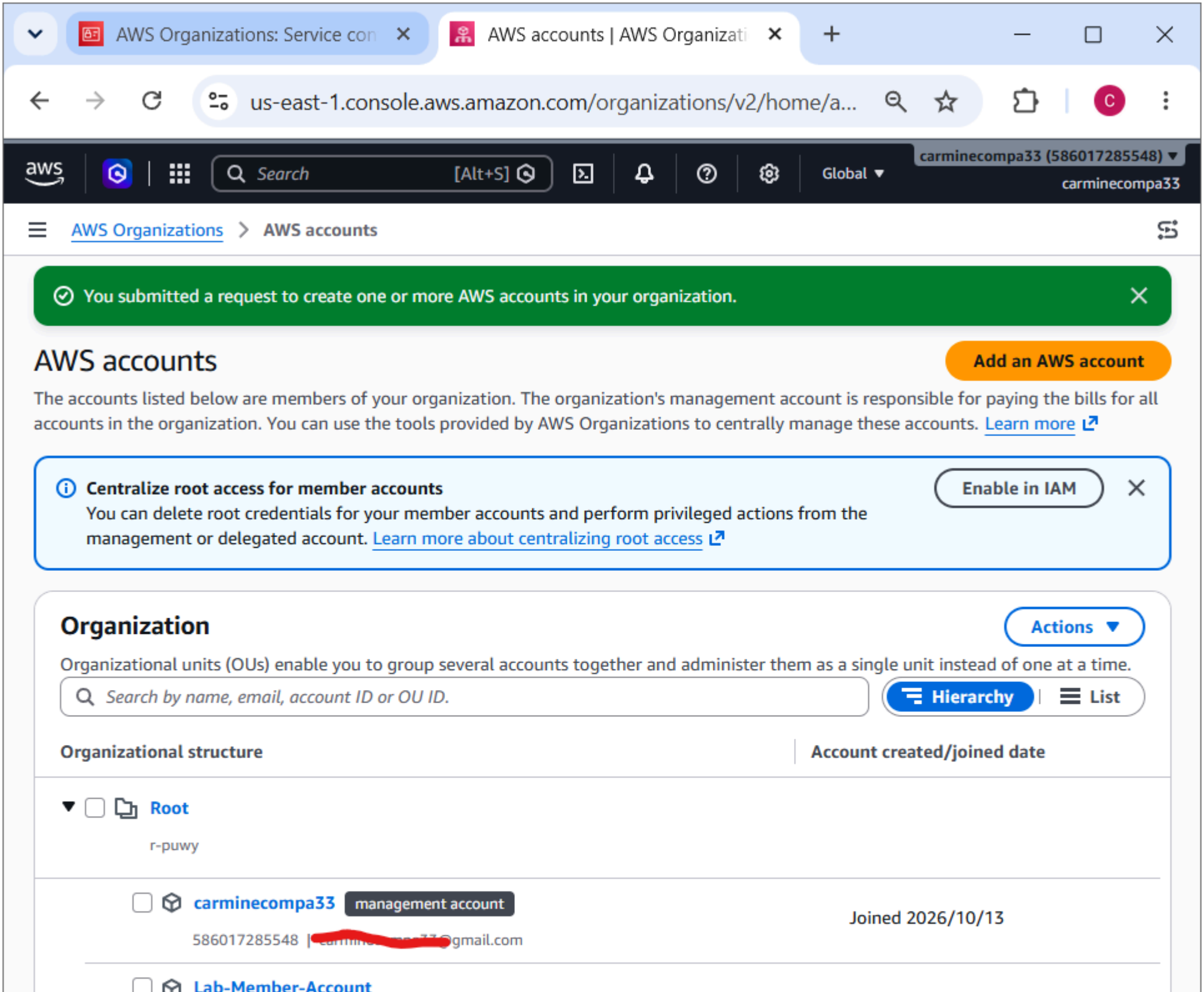
Copy

```
            "cloudfront:*",
            "sts:*",
            "support:*",
            "securityhub:*",
            "config:*"
        ],
        "Resource": "*",
        "Condition": {
            "StringNotEquals": {
                "aws:RequestedRegion": [
                    "us-east-2",
                    "us-east-1"
                ]
```

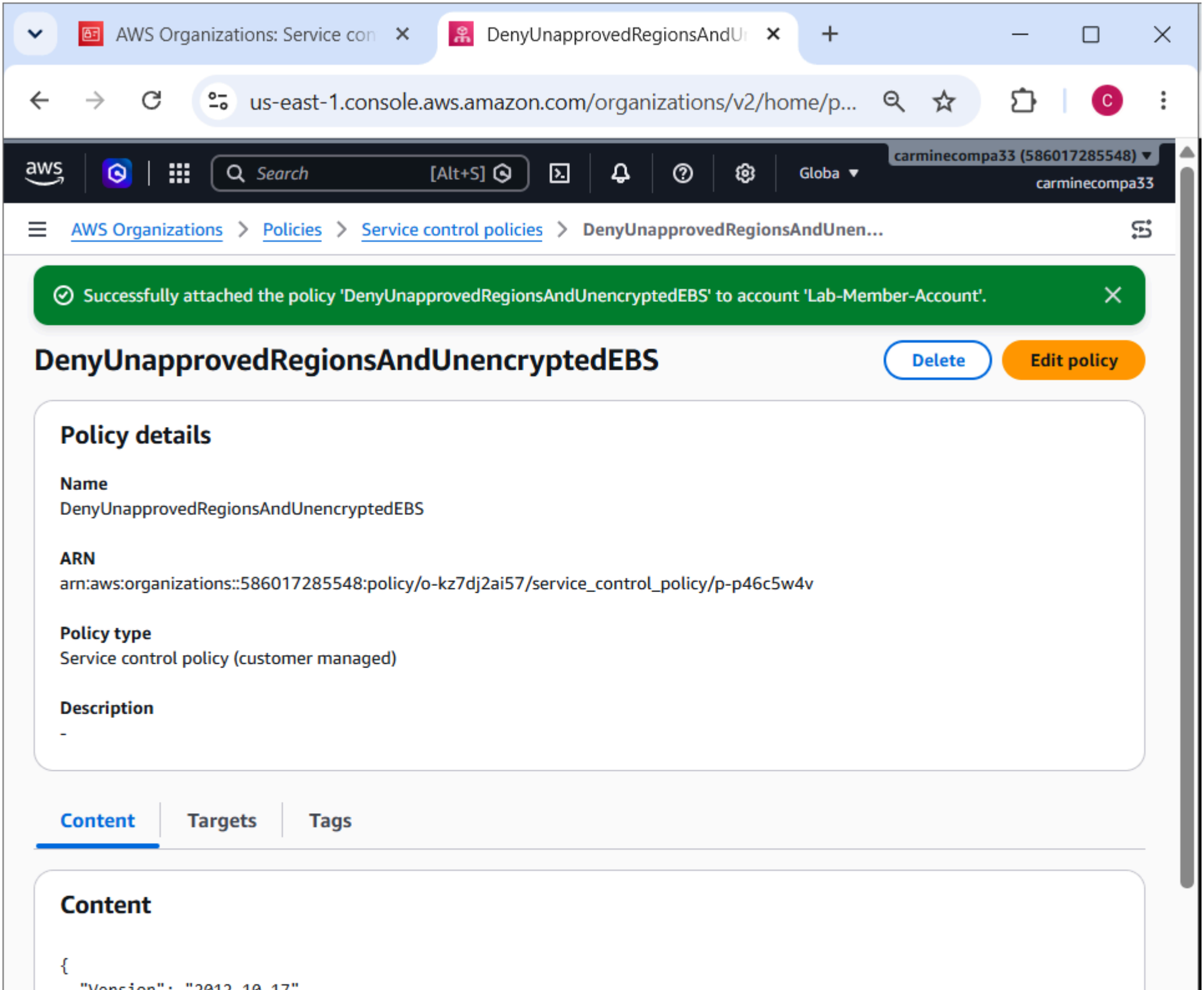
However I don’t have Member Accounts created yet


AWS Organizations: Service con
AWS accounts | AWS Organizati
us-east-1.console.aws.amazon.com/organizations/v2/home/a...
Search
[Alt+S]
Global
carminecompa33 (586017285548)
carminecompa33
AWS Organizations
AWS accounts
AWS accounts
Add an AWS account
The accounts listed below are members of your organization. The organization's management account is responsible for paying the bills for all accounts in the organization. You can use the tools provided by AWS Organizations to centrally manage these accounts. Learn more
Centralize root access for member accounts
You can delete root credentials for your member accounts and perform privileged actions from the management or delegated account. Learn more about centralizing root access
Enable in IAM
Organization
Actions
Organizational units (OUs) enable you to group several accounts together and administer them as a single unit instead of one at a time.
Search by name, email, account ID or OU ID.
Hierarchy
List
Organizational structure
Account created/joined date
Root
r-puwy
carminecompa33
management account
586017285548 | carminecompa33@gmail.com
Joined 2026/10/13

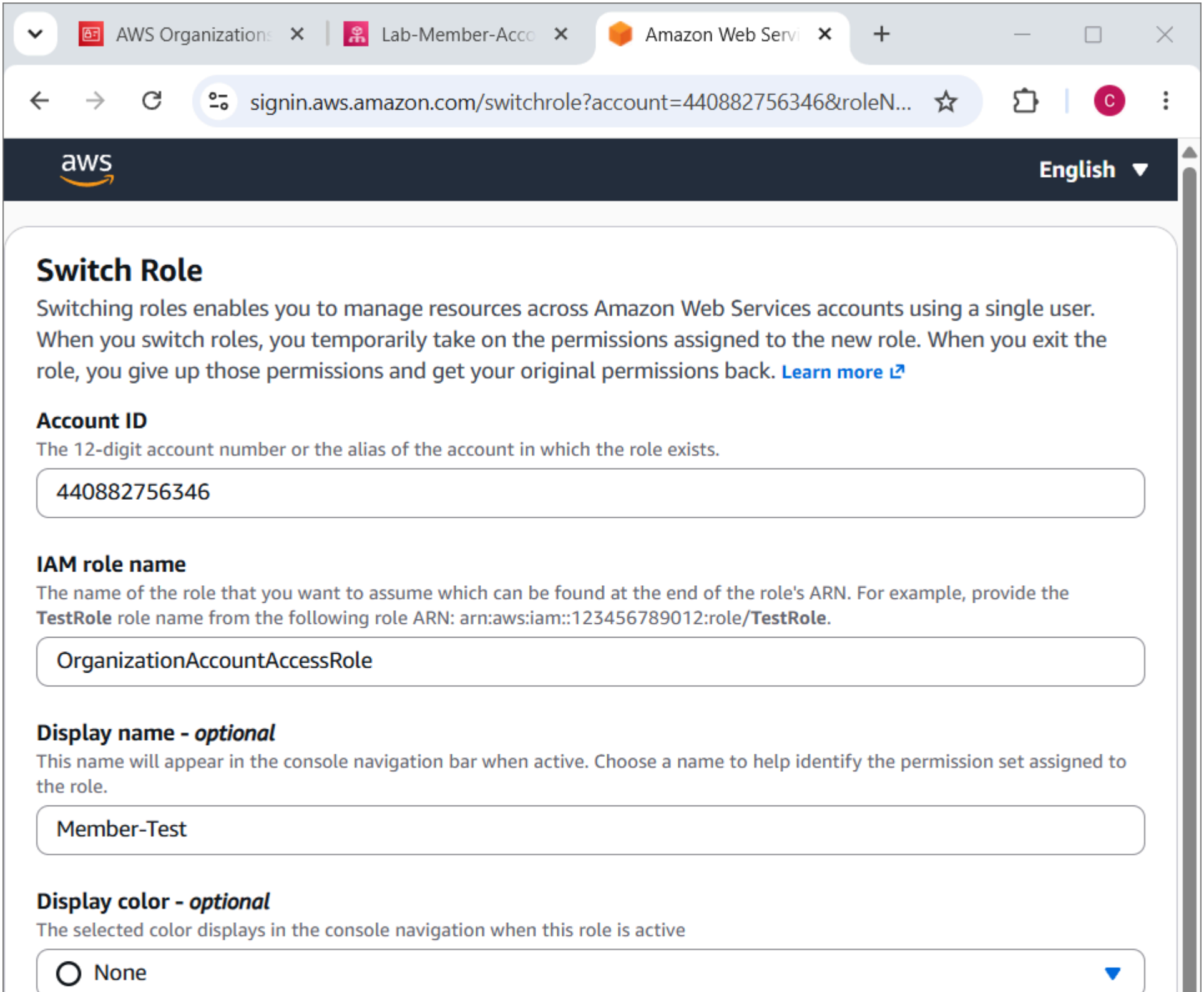
Created a new Member Account


AWS Organizations: Service con
AWS accounts | AWS Organizati
us-east-1.console.aws.amazon.com/organizations/v2/home/a...
aws
Search
[Alt+S]
Global
carminecompa33 (586017285548)
carminecompa33
AWS Organizations
AWS accounts
You submitted a request to create one or more AWS accounts in your organization.
AWS accounts
Add an AWS account
The accounts listed below are members of your organization. The organization's management account is responsible for paying the bills for all accounts in the organization. You can use the tools provided by AWS Organizations to centrally manage these accounts. Learn more
Centralize root access for member accounts
You can delete root credentials for your member accounts and perform privileged actions from the management or delegated account. Learn more about centralizing root access
Enable in IAM
Organization
Actions
Organizational units (OUs) enable you to group several accounts together and administer them as a single unit instead of one at a time.
Search by name, email, account ID or OU ID.
Hierarchy
List
Organizational structure
Account created/joined date
Root
r-puwy
carminecompa33
management account
586017285548
gmail.com
Joined 2026/10/13
Lab-Member-Account

Attached the new Member Account to the policy DenyUnapprovedRegionsAndUnencryptedEBS


AWS Organizations: Service con
DenyUnapprovedRegionsAndU
us-east-1.console.aws.amazon.com/organizations/v2/home/p...
aws
Search
[Alt+S]
Globa
carminecompa33 (586017285548)
carminecompa33
AWS Organizations
Policies
Service control policies
DenyUnapprovedRegionsAndUnen...
Successfully attached the policy 'DenyUnapprovedRegionsAndUnencryptedEBS' to account 'Lab-Member-Account'.
DenyUnapprovedRegionsAndUnencryptedEBS
Delete
Edit policy
Policy details
Name
DenyUnapprovedRegionsAndUnencryptedEBS
ARN
arn:aws:organizations::586017285548:policy/o-kz7dj2ai57/service_control_policy/p-p46c5w4v
Policy type
Service control policy (customer managed)
Description
-
Content
Targets
Tags
Content
{

Switching role to the new created Member Account Member-Test


AWS Organizations
Lab-Member-Acco
Amazon Web Servi
signin.aws.amazon.com/switchrole?account=440882756346&roleN...
aws
English
Switch Role
Switching roles enables you to manage resources across Amazon Web Services accounts using a single user. When you switch roles, you temporarily take on the permissions assigned to the new role. When you exit the role, you give up those permissions and get your original permissions back. Learn more
Account ID
The 12-digit account number or the alias of the account in which the role exists.
440882756346
IAM role name
The name of the role that you want to assume which can be found at the end of the role's ARN. For example, provide the TestRole role name from the following role ARN: arn:aws:iam::123456789012:role/TestRole.
OrganizationAccountAccessRole
Display name - optional
This name will appear in the console navigation bar when active. Choose a name to help identify the permission set assigned to the role.
Member-Test
Display color - optional
The selected color displays in the console navigation when this role is active
None

The system doesn’t allow me to switch role

signin.aws.amazon.com/switchrole?account=440882756346&roleN...

aws

English

Switching roles enables you to manage resources across Amazon Web Services accounts using a single user. When you switch roles, you temporarily take on the permissions assigned to the new role. When you exit the role, you give up those permissions and get your original permissions back. Learn more

Account ID

The 12-digit account number or the alias of the account in which the role exists.

440882756346

IAM role name

The name of the role that you want to assume which can be found at the end of the role's ARN. For example, provide the **TestRole** role name from the following role ARN: arn:aws:iam::123456789012:role/**TestRole**.

OrganizationAccountAccessRole

Display name - *optional*

This name will appear in the console navigation bar when active. Choose a name to help identify the permission set assigned to the role.

Member-Test

Display color - *optional*

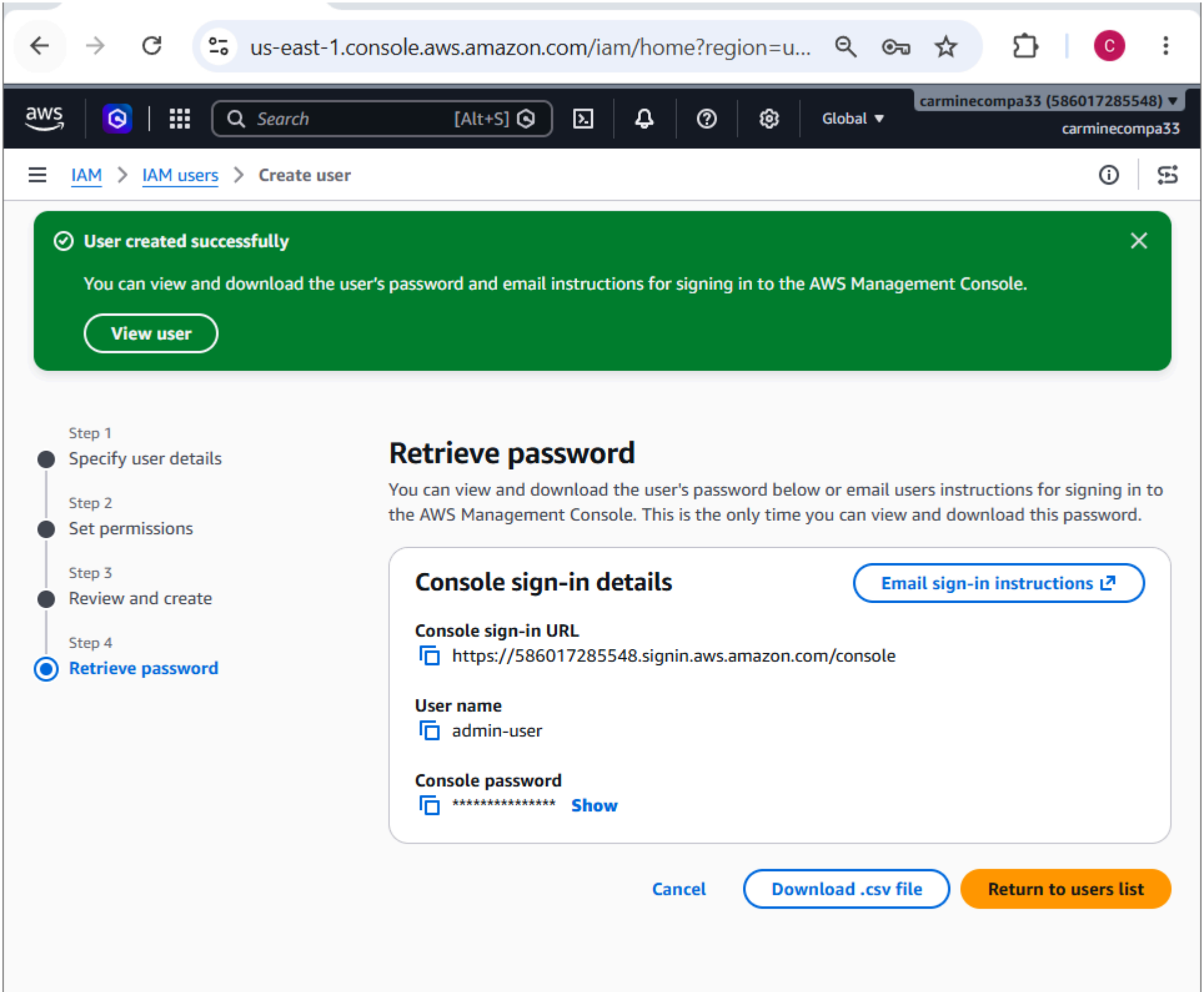
The selected color displays in the console navigation when this role is active

None

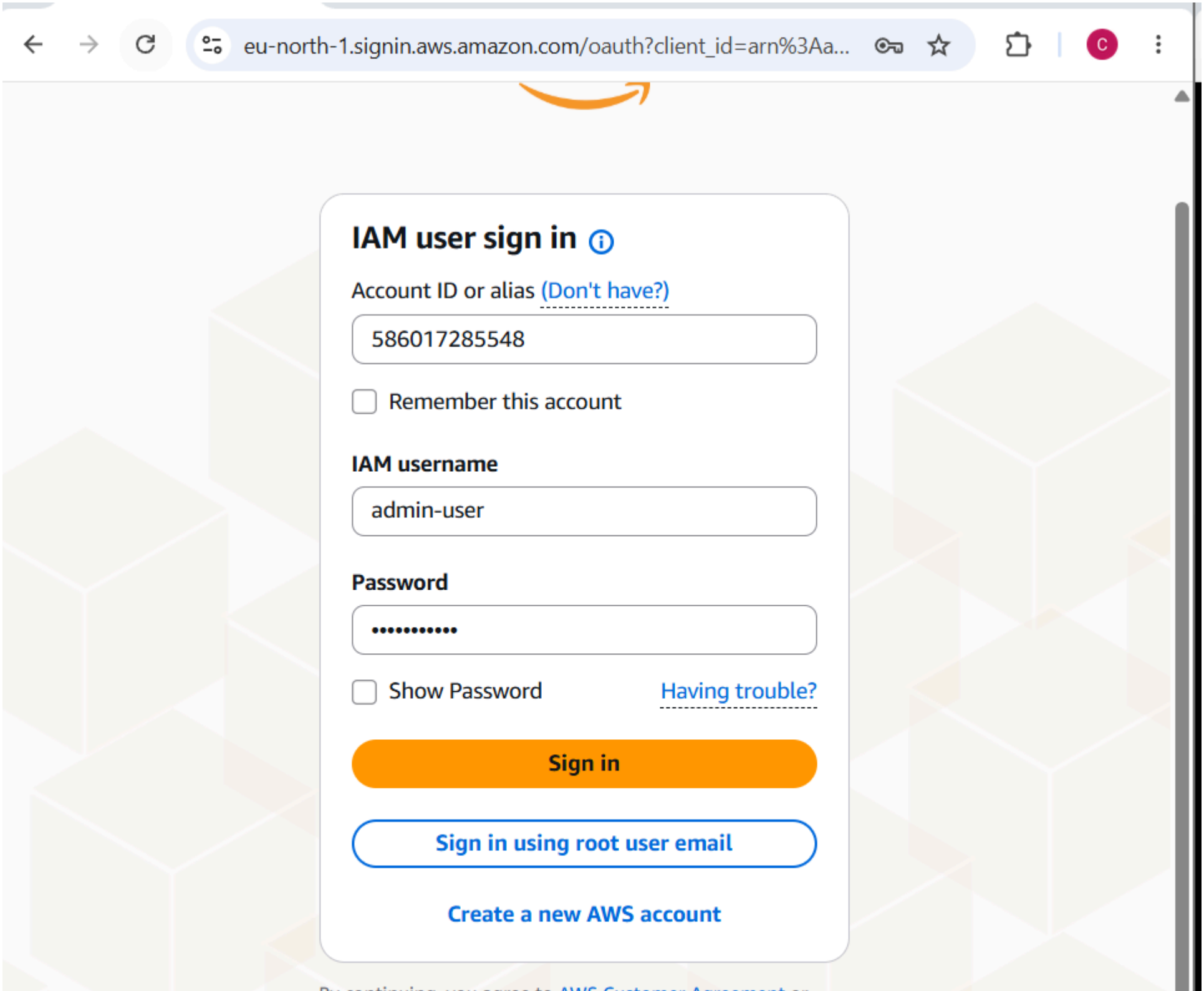
Invalid information in one or more fields

Probably I need to adopt a IAM Role

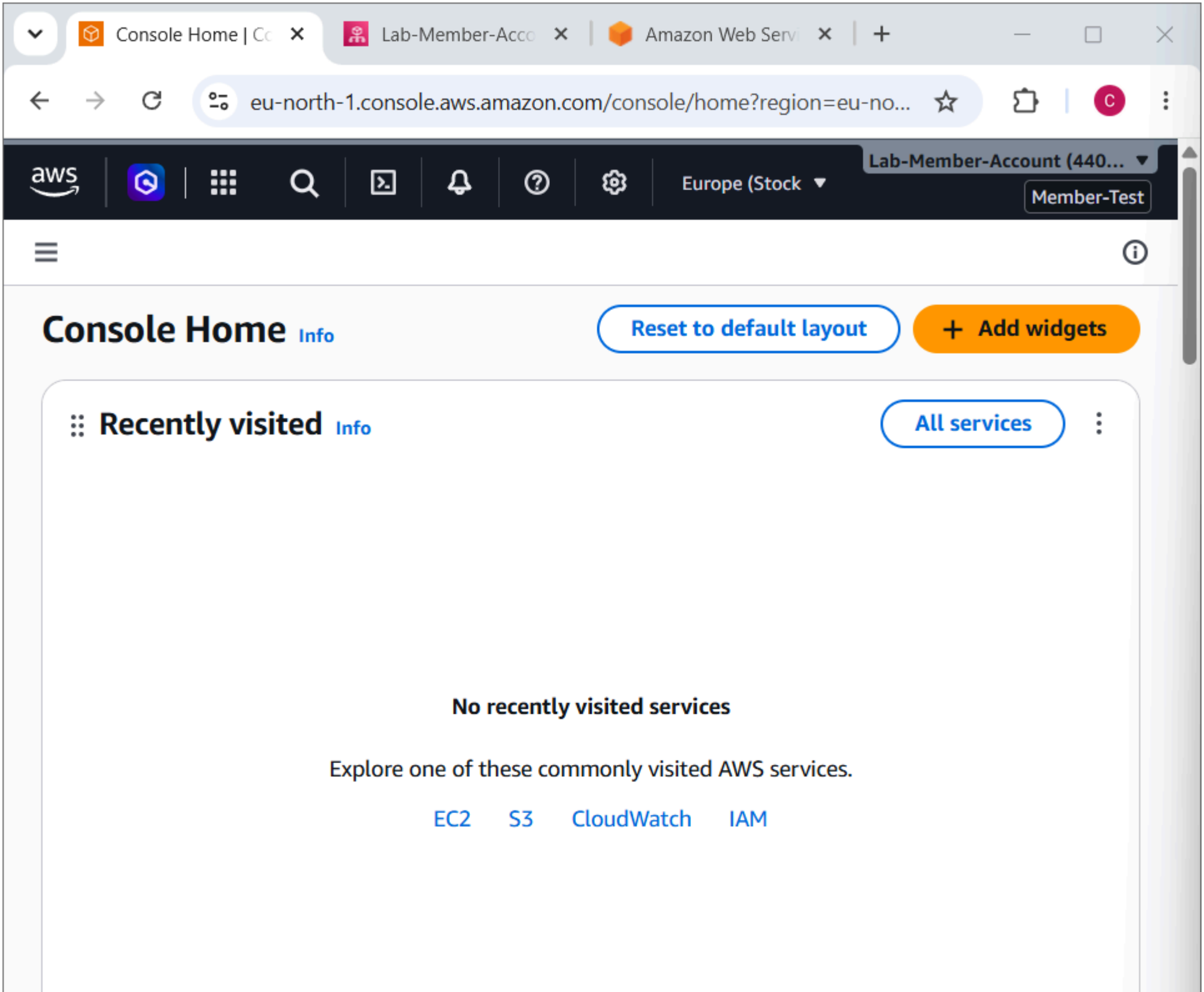
Created a IAM User with AdministratorAccess policy


us-east-1.console.aws.amazon.com/iam/home?region=u...
aws
Search
[Alt+S]
Global
carminecompa33 (586017285548)
carminecompa33
IAM
IAM users
Create user
User created successfully
You can view and download the user's password and email instructions for signing in to the AWS Management Console.
View user
Step 1
Specify user details
Step 2
Set permissions
Step 3
Review and create
Step 4
Retrieve password
Retrieve password
You can view and download the user's password below or email users instructions for signing in to the AWS Management Console. This is the only time you can view and download this password.
Console sign-in details
Email sign-in instructions
Console sign-in URL
https://586017285548.signin.aws.amazon.com/console
User name
admin-user
Console password
****
Show
Cancel
Download .csv file
Return to users list

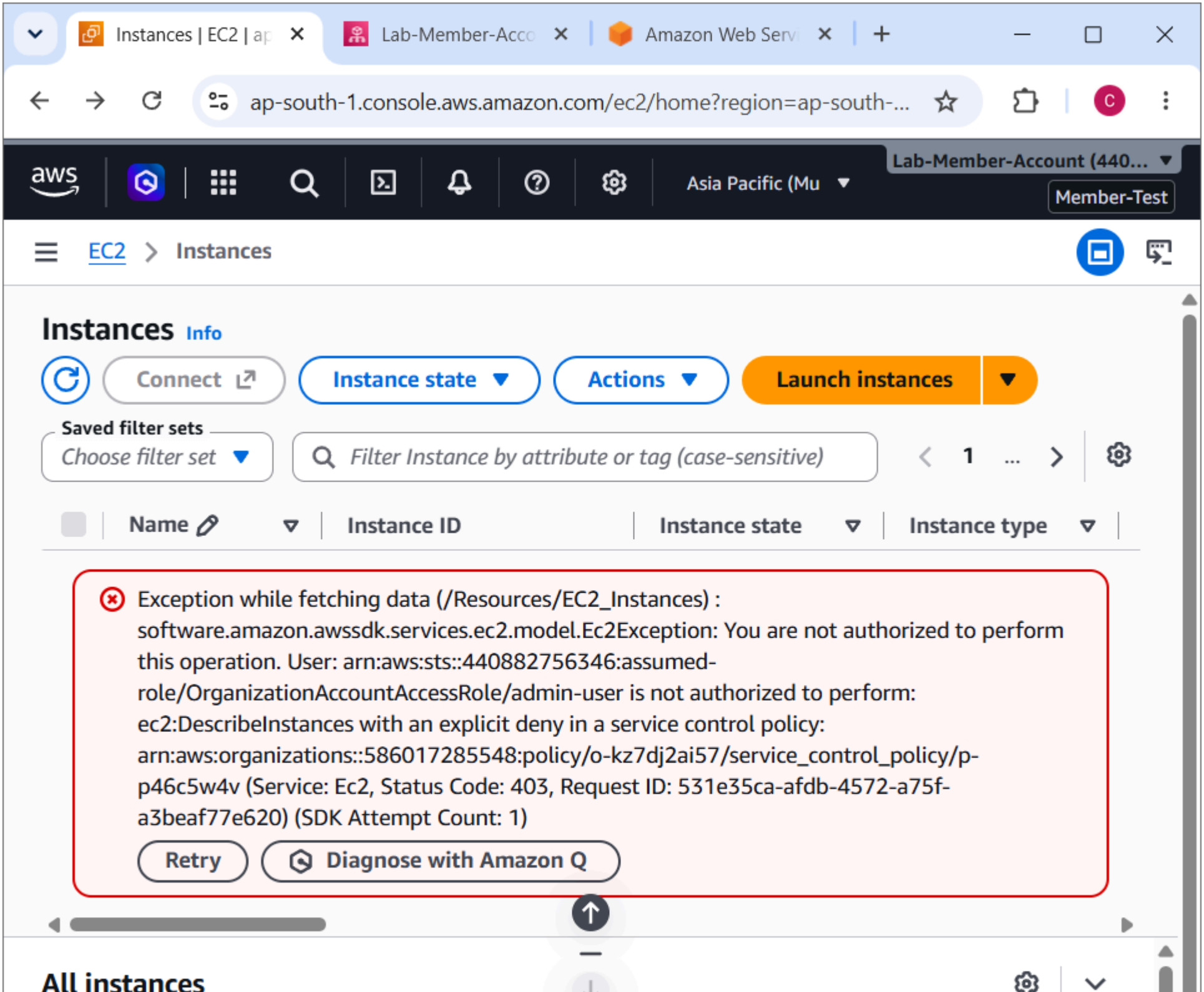
Log Out from my root org user
Login with admin-user


eu-north-1.signin.aws.amazon.com/oauth?client_id=arn%3Aa...
IAM user sign in
Account ID or alias (Don't have?)
586017285548
Remember this account
IAM username
admin-user
Password
Show Password
Having trouble?
Sign in
Sign in using root user email
Create a new AWS account

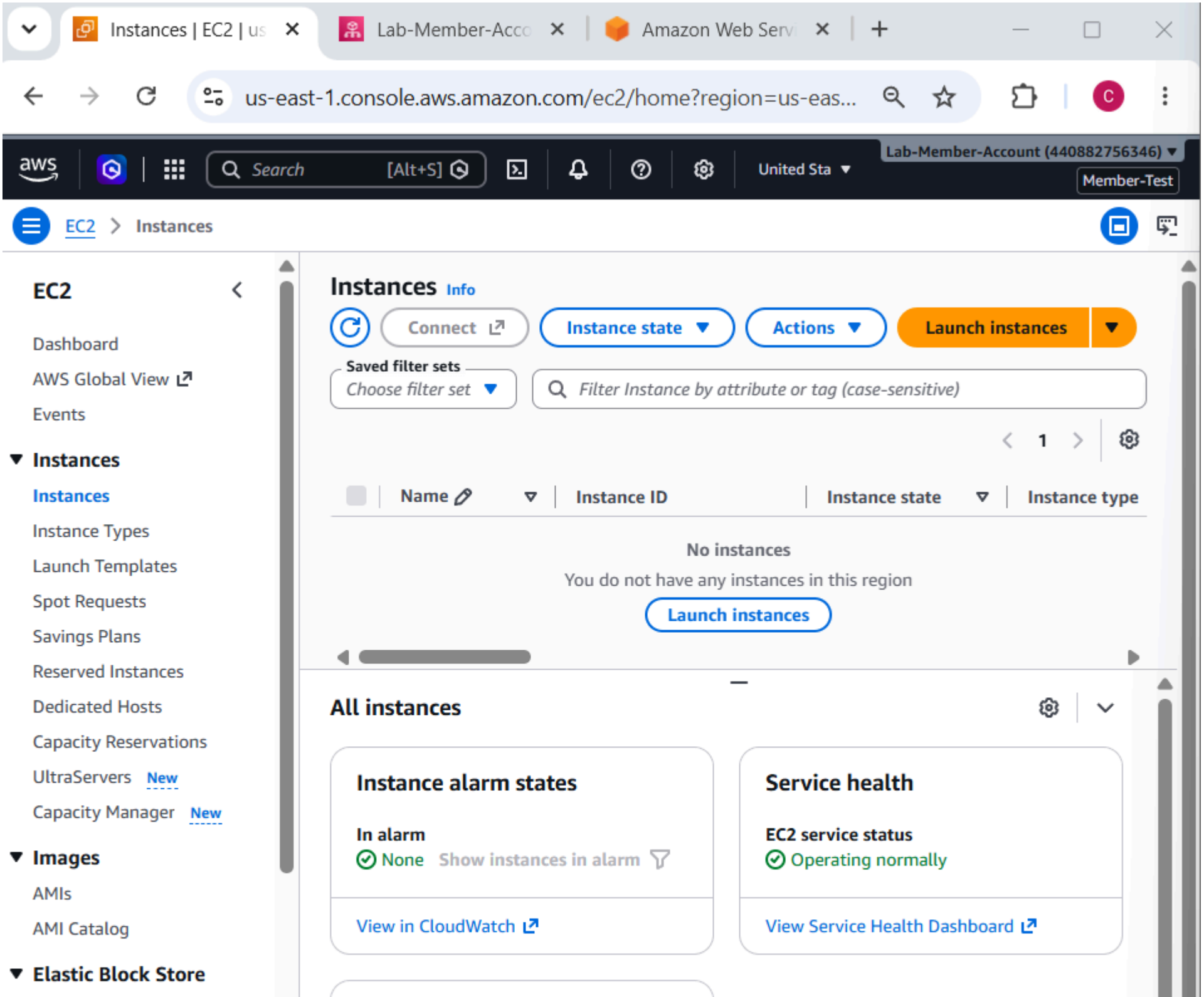
Now I can switch to the Member-Test account


Console Home | Cc
Lab-Member-Acco
Amazon Web Servi
eu-north-1.console.aws.amazon.com/console/home?region=eu-no...
aws
Europe (Stock
Lab-Member-Account (440...
Member-Test
Console Home Info
Reset to default layout
+ Add widgets
Recently visited Info
All services
No recently visited services
Explore one of these commonly visited AWS services.
EC2
S3
CloudWatch
IAM

SCP policy is working: with Member-Account I cannot create resources in Asia Pacific Mumbai


Instances | EC2 | ap
Lab-Member-Acco
Amazon Web Servi
ap-south-1.console.aws.amazon.com/ec2/home?region=ap-south-...
aws
Asia Pacific (Mu
Lab-Member-Account (440...
Member-Test
EC2 > Instances
Instances Info
Connect
Instance state
Actions
Launch instances
Saved filter sets
Choose filter set
Filter Instance by attribute or tag (case-sensitive)
1 ...
Name
Instance ID
Instance state
Instance type
Exception while fetching data (/Resources/EC2_Instances) : software.amazon.awssdk.services.ec2.model.Ec2Exception: You are not authorized to perform this operation. User: arn:aws:sts::440882756346:assumed-role/OrganizationAccountAccessRole/admin-user is not authorized to perform: ec2:DescribeInstances with an explicit deny in a service control policy: arn:aws:organizations::586017285548:policy/o-kz7dj2ai57/service_control_policy/p-p46c5w4v (Service: Ec2, Status Code: 403, Request ID: 531e35ca-afdb-4572-a75f-a3beaf77e620) (SDK Attempt Count: 1)
Retry
Diagnose with Amazon Q
All instances

But I can us-east1


Instances | EC2 | us
Lab-Member-Acco
Amazon Web Serv
us-east-1.console.aws.amazon.com/ec2/home?region=us-eas...
Search [Alt+S]
United Sta
Lab-Member-Account (440882756346)
Member-Test
EC2 > Instances
EC2
Dashboard
AWS Global View
Events
Instances
Instances
Instance Types
Launch Templates
Spot Requests
Savings Plans
Reserved Instances
Dedicated Hosts
Capacity Reservations
UltraServers New
Capacity Manager New
Images
AMIs
AMI Catalog
Elastic Block Store
Instances Info
Connect
Instance state
Actions
Launch instances
Saved filter sets
Choose filter set
Filter Instance by attribute or tag (case-sensitive)
1
Name
Instance ID
Instance state
Instance type
No instances
You do not have any instances in this region
Launch instances
All instances
Instance alarm states
In alarm
None
Show instances in alarm
View in CloudWatch
Service health
EC2 service status
Operating normally
View Service Health Dashboard

Enabled AWS Config

Dashboard | AWS Config | us-ea

us-east-2.console.aws.amazon.com/config/home?region=us-...

aws | Search [Alt+S] | United Sta | carminecompa33 (586017285548) | carminecompa33

AWS Config > Dashboard

AWS Config

- Dashboard
- Conformance packs
- Rules
- Resources
- Aggregators
  - Compliance Dashboard
  - Conformance packs
  - Rules
  - Inventory Dashboard
  - Resources
  - Authorizations
- Advanced queries
- Settings

- Documentation
- Partners
- FAQs
- Pricing

All resources

Type	Count

Your resources are being discovered

Run advanced queries against your resource configuration data.

AWS Config usage metrics

AWS Config usage metrics by resource type

Choose Resource types

All

3h 1d 1w | UTC timezone | Explore related

Configuration Items ...

No data available.

Try adjusting the dashboard time range.

Configuration Recor...

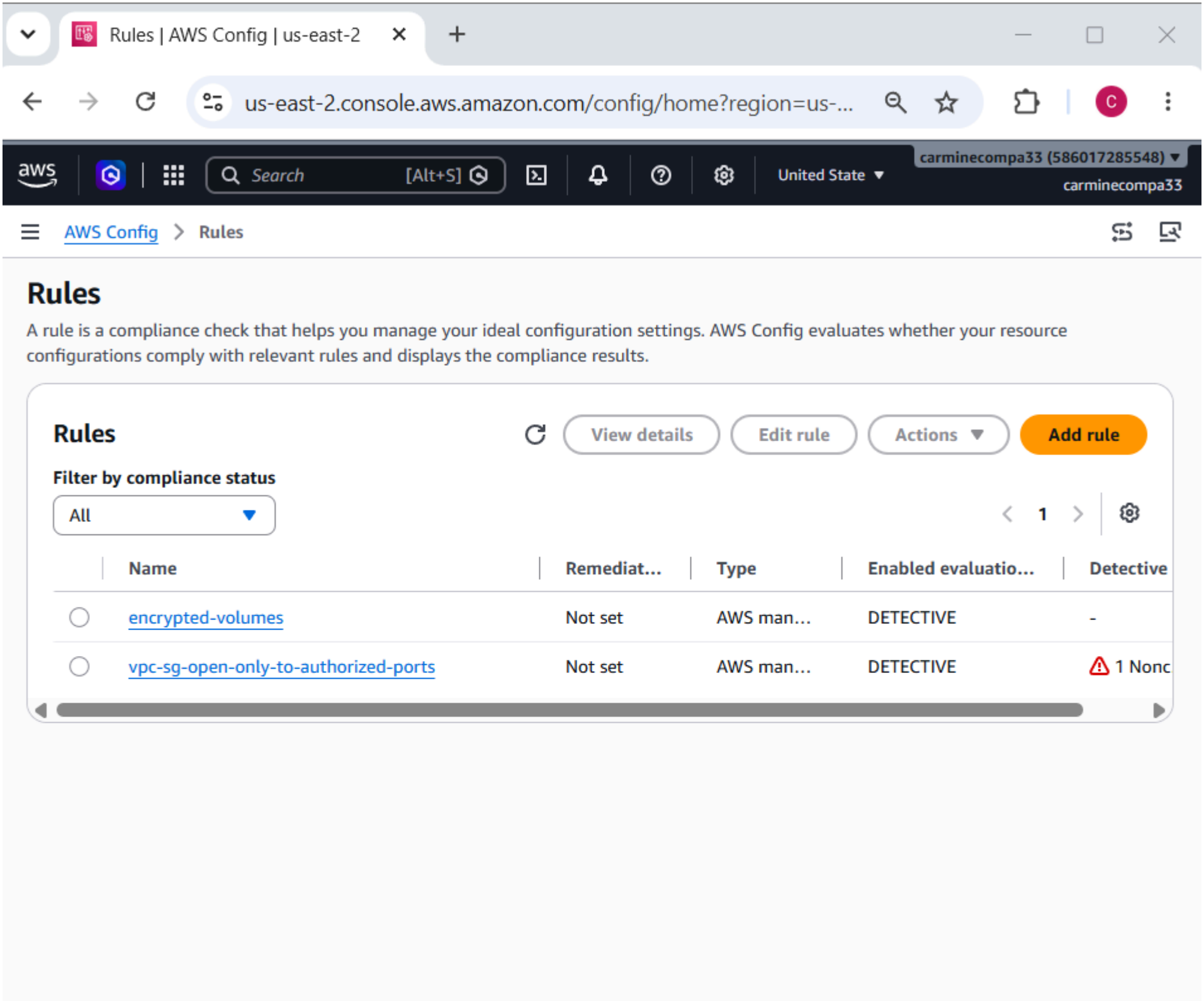
Count

1

0.5

AWS Config has 2 rule

- encrypted-volumes
- vpc-sg-open-only-to-authorized-ports


Rules | AWS Config | us-east-2
us-east-2.console.aws.amazon.com/config/home?region=us-...
aws
Search
[Alt+S]
United State
carminecompa33 (586017285548)
carminecompa33
AWS Config
Rules
Rules
A rule is a compliance check that helps you manage your ideal configuration settings. AWS Config evaluates whether your resource configurations comply with relevant rules and displays the compliance results.
Rules
View details
Edit rule
Actions
Add rule
Filter by compliance status
All
1
Name
Remediat...
Type
Enabled evaluatio...
Detective
encrypted-volumes
Not set
AWS man...
DETECTIVE
-
vpc-sg-open-only-to-authorized-ports
Not set
AWS man...
DETECTIVE
1 Nonc

Enabled CIS AWS Foundations Benchmark 1.4.0 from Security Hub

Posture management | Security

Security standards | Security Hu

us-east-2.console.aws.amazon.com/securityhub/v2/home?reg...

aws | Search [Alt+S] | United Sta | carminecompa33 (586017285548) | carminecompa33

Security Hub > Posture management

Choose a filter set | Add filter

Status = New | or | Status = In Progress | or | Status = Other

or | Status = Unknown

and | Provider Status != Close | and | Compliance status != Pass | Clear filter

Dashboard update

Your dashboard has been updated with new views and widgets. You can customize the views by adding or removing the widgets.

Security standards

All filters not applied

View the compliance status of your enabled security standards. The score represents the percentage of passed checks for each standard.

Standard	Passed	Failed	Score
AWS Foundational Security Best Practices v1.0.0	0	0	0%
CIS AWS Foundations Benchmark v1.2.0	0	0	0%
AI Security Best Practices v1.0.0			Enable
AWS Resource Tagging Standard v1.0.0			Enable

Posture management | Security

Security standards | Security Hu

us-east-2.console.aws.amazon.com/securityhub/home?region...

aws | Search [Alt+S] | United Sta | carminecompa33 (586017285548) | carminecompa33

Security Hub CSPM > Security standards

Successfully enabled standard: CIS AWS Foundations Benchmark v1.4.0

Provider
Azure

Enable standard

Enable standard

CIS AWS Foundations Benchmark v1.2.0

Description
The Center for Internet Security (CIS) AWS Foundations Benchmark v1.2.0 is a set of security configuration best practices for AWS. This Security Hub standard automatically checks for your compliance readiness against a subset of CIS requirements.

Created by
AWS Security Hub

Provider
AWS

Security score

Unable to display security score: The security

CIS AWS Foundations Benchmark v1.4.0

Description
The Center for Internet Security (CIS) AWS Foundations Benchmark v1.4.0 is a set of security configuration best practices for AWS. This Security Hub standard automatically checks for your compliance readiness against a subset of CIS requirements.

Created by
AWS Security Hub

Provider
AWS

Security score

Unable to display security score: The security